First we'd open our Q1 icon. Next, we'd open our AppleWorks folder, then tunnel into the AW.Files folder (hold down option while double-clicking). (We won't be doing anything with the AppleWorks folder itself; we only opened it to find our AW.Files folder, so we automatically closed it by tunneling.) We position the windows so that we can see both the files we want to move (or at least some of them) and the folder we want to move them into.

Now we select the AppleWorks document icons. We may have to scroll the window with the scroll bars to find all the files we want to move; we'll select those by shift-clicking them. Once they're all selected, we drag any one of them to the AW.Files folder, making sure that the AW.Files folder icon is blackened before we release the mouse button. When we release the mouse button, the files are moved.

Filename Translation

When you copy a file from one type of disk to another, and the name of the source file doesn't follow the rules of the target disk (for example, if you're copying a file from a Macintosh disk to a ProDOS disk and the file's name is longer than 15 characters), the Finder presents a dialog (Figure 6K) that allows you to specify a new name for the file. The Finder tries to turn the original file's name into a name that will fit the ProDOS disk, then offers that to you as a default name. You can edit the name before saving the file (or you can skip copying the file entirely).

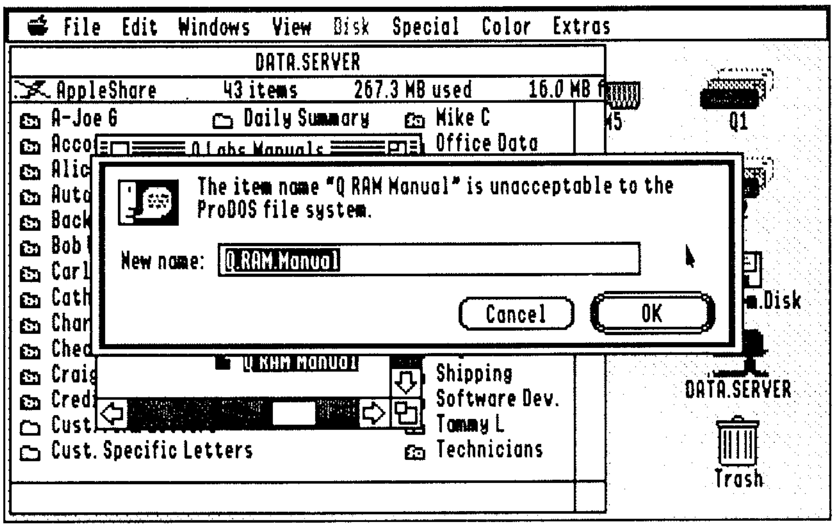


Figure 6K—Filename Translation Dialog

If you want the Finder to automatically translate the names of the rest of the files you're copying, click the "Translate bad names" button. If you want it to automatically skip any files which have bad names, click the "Skip bad names" button.

▲ **By The Way...**

You'll get a similar message in other applications if you try to save a document to disk with a name that doesn't fit the naming rules of the disk you're saving it on.

Copying On The Same Disk

When we drag files to another directory on the same disk, the Finder assumes we want to move them (instead of making copies) because most of the time, we don't need or want two copies of the same file on our disks. When we actually do want to make a copy of a file on the same disk, we have a couple of alternatives. Unfortunately, none of them are particularly elegant, but they'll have to do. Thankfully, it's probably not something you'll need to do very often.

First is the File menu's "Duplicate..." command. Select the icon you want to make a copy of, then choose this option (or press ⌘-D). The Finder will present you with a dialog (Figure 6L) which allows you to name the new copy of the file. (You can't use the same name as the original because the duplicate copy will be placed into the same window as the original file.) Type a new name or leave the Finder's default name, and click "OK" (or press Return). Now you can drag the duplicate file to its final resting place. If it's important that the new file has the same name as the original, you can rename it after it's been moved.

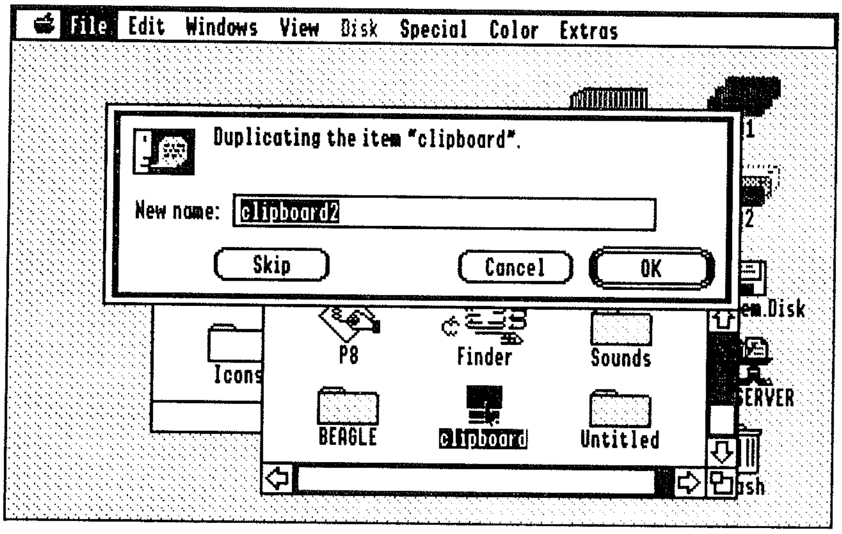


Figure 6L—Duplicate (Make A Copy) Dialog

If you want to copy a group of files that are in a folder, it may be more convenient to duplicate the *folder the items are in* instead of the individual items. Use the pop-up folder menu to bring the folder's parent window to the front, then select the folder and choose "Duplicate..." from the File menu (or, again, press ⌘-D). Click "OK" (or press Return) to accept the name for the new folder. Now you've got two copies of the folder; the contents of both folders have identical names. (The files can have the same names because they're in different folders, but the folders have different names.) Now you can move the duplicated files—or the whole folder.

A final way to make a copy of a file on the same disk is to simply copy the file to another disk, then back to the original disk, and finally delete the temporary copy from the second disk. Often this is the long way around, but if your system contains a RAM Disk, you may actually find it the fastest and most convenient way.

The Trash

The Trash is used to remove files and folders from your disks. Think of the trash as a holding place for things you want to get rid of. The items aren't deleted immediately, but are instead held in the trash for a while to give you a chance to change your mind. When the trash has something in it, it bulges (Figure 6M).

Figure 6M—The Bulging Trash Can

The trash is emptied automatically when you copy or move a file, when you launch a program or open a document, or when you shut down using the "Shut Down" option on the Special menu. You can also empty it manually with the "Empty Trash" option under Special.

If you decide you don't want to delete a file you put in the Trash, you can open the Trash like any other icon by double-clicking it. Select the items you changed your mind about and select "Put Away" from the File menu. The files will return to their original folders and disks. Alternately, you can drag them to where they belong if you don't want to return them to their original locations.

☞ **Note!**

If you turn off the computer's power before the Finder empties the trash (i.e., if you don't use the "Shut Down" option), the contents of the trash return to their original windows.

Files On The Desktop

It's possible to drag files *out* of their windows and deposit them on the Finder's desktop. The Finder will remember the positions and coloring of these icons, so they'll stay there until you want them to go away. By doing this with frequently-used files and folders, you can streamline how you work with your disks in the Finder.

Prime candidates for desktop icons include your most frequently used programs, like AppleWorks or AppleWorks GS, plus any other programs you use more than occasionally. Placing a folder on the desktop allows easy access to the entire contents of the folder with just a double-click. By placing files and folders on the desktop, you can turn the Finder into a customized program launcher that rivals most commercial packages.

Since the Finder places your disk icons on the right side of the screen, most users place their other desktop icons on the left side of the screen or along the bottom.

☞ **Note!**

When you move an icon to the desktop, it doesn't really go anywhere. The Finder doesn't show it in the window it used to be in, but most other programs will. If you move a file or folder to the desktop and need to find it from another program, you'll find it in the same directory it was in before you moved it to the desktop.

▲ **By The Way...**

The "Put Away" option on the File menu works on desktop files, too. Click the icons you don't want on your desktop anymore and choose "Put Away," and the icons will move back to the directory from whence they came. (You can, of course, still move the files manually.) "Clean Up" (Special menu) can be used to align the icons on the desktop to a grid; just close all open windows ("Close All" under File) first.

Working With Programs & Documents

Opening Programs

To open, or launch, an application, find the program's icon (usually a diamond icon with a hand writing on it—see Figure 6N) and double-click it.

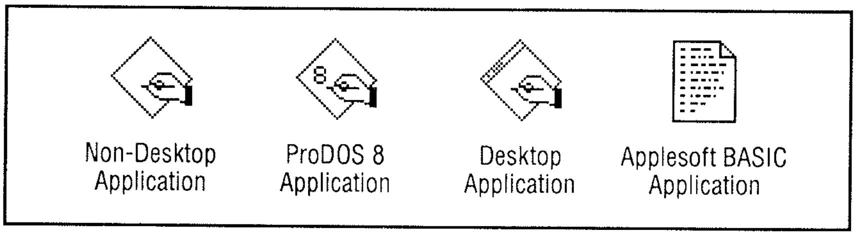


Figure 6N—Standard System 6 Icons for Applications

System 6 supports several different types of application programs. To remain compatible with older Apple II models, System 6 can run 8-bit SYS applications such as AppleWorks. These programs are identified by the hand-writing-on-paper icon with a numeral "8" in it. Many 8-bit applications have names that end in the word "System."

System 6 can also run Applesoft BASIC programs directly from the Finder. These programs usually have an icon which looks like a regular document with several lines of rainbow-colored writing on them. Applesoft applications frequently have the name "Startup." Some BASIC programs will return to the Finder on their own; if you find yourself looking at an Applesoft prompt (]) when the program has finished running, type the word BYE and press Return to go back to the Finder. (You can also use the BYE command if you accidentally run BASIC.System.)

▲ By The Way...

Applesoft BASIC programs are treated as documents by the Finder—that's why they have the folded-down corner. The documents "belong" to the program named BASIC.System. See the next section for more on documents and how they can "belong" to application program.

Of course, System 6 can also run modern Desktop-based programs like AppleWorks GS. Icons for desktop programs have a tiny menu bar. Many such programs have names that end in "Sys16." There are also 16-bit programs that *don't* use the Desktop; they're indicated by the generic application icon—the plain diamond with the hand writing on it.

☞ Note!

When you run an 8-bit SYS application, an Applesoft BASIC program, or a non-Desktop 16-bit program, you leave the Apple Desktop environment. Some programs mimic the Apple Desktop in part or in whole, but in general, each non-Desktop program is different.

Opening Documents

One of the Finder's most powerful features is its ability to open document files in the program that created them. If you double-click a document icon, the Finder will open it if it knows which program it "belongs" to. That is, the Finder will run the program that the document belongs to and tell it to open your document. (If the Finder doesn't know which program owns a document, it will display an "An Application Can't Be Found For This Document" alert.)

As shipped from Apple, the Finder knows only that BASIC programs belong to a program called BASIC.System, and that standard text files belong to a simple word processor (included with System 6) called Teach.

You tell the Finder about what documents go with which programs via icon files. Icon files contain special icons for the files that are associated with a program. They also contain the information the Finder needs to "link" the program and its documents. Many programs come with icon files; when you install such programs on your hard drive, you should also copy the program's icon file into the Icons folder on your hard drive.

⑥ New & Improved

Under System 6, it's possible for a programmer to include the icons within the application program. In that case, copying the program to your hard drive is all that's necessary—new icons for the program and its documents will appear automatically. More and more new programs will use this technique, making icon files a thing of the past.

Sometimes the Finder will know that a document goes with a particular program, but will be unable to find the program because the program isn't in the folder that the Finder is looking in, or because the disk the program is on isn't a drive. When this happens, the Finder will alert you that it can't find the program and permit you to Cancel, Try Again (after you've inserted the disk that contains the program), or Locate. If you choose Locate, a Standard File dialog appears so you can show the Finder where the program is. (We'll cover Standard File dialogs in the next chapter.) Once you've located a program for the Finder, the Finder remembers where the program is so that it knows how to open that kind of document in the future.

The Finder also allows you to override a particular document-application link temporarily by holding down the Option key while you double-click the document. The Finder will present a Standard File dialog to allow you to open the document with a different application. The link will be temporary—only the selected documents will be opened using the application you choose. To make a new link permanent, hold down *both* Option and Control while you double-click a document; the Finder will forget the old application and remember that the new application should be used to open all documents of that type.

☞ **Note!**

The Option and Control-Option features only work when there's already a link in place for a specific kind of file. For example, Option-opening a font has no effect if the Finder doesn't already know that fonts should be opened with a particular application. (More specifically, the Finder doesn't create new links for unlinked documents, only update existing links.) To create links of your own, you need an application called an icon editor. Two popular icon editors are IconEd by Paul Elseth and DIcEd by Dave Lyons; both are shareware and are available from user groups, online services, and public-domain software distributors.

☞ **Note!**

Basic Launcher

If you previously used System 5, the Finder may occasionally ask for an application called BASIC.Launcher when you try to launch BASIC programs. (Remember, the Finder considers BASIC programs to be documents for the BASIC application.) The BASIC.Launcher application is no longer present (or necessary) in System 6. Use the Locate button to re-link the BASIC program to BASIC.System. (Alternately, you can Duplicate the BASIC.System file and name the copy BASIC.Launcher; then the Finder will always be able to find BASIC.Launcher.)

needed for copy II+

The Finder can also tell the application to print your document automatically. Just select the document(s), then select "Print" from the File menu. The Finder will launch the application and tell it to open your documents and print them.

☞ **Note!**

Some programs don't support automatically opening document files. Others support opening document files, but can't automatically print them. The SynthLab program that comes with System 6 is one of the former; the Teach mini-word processor is one of the latter.

Other Finder Features

The Finder has several features we haven't discussed yet. We'll cover them in order, from left to right across the menu bar. Many of these features are quite useful, but they didn't fit in very neatly with our tutorial-style Finder instructions.

About The Finder (Menu)

The "About the Finder" option opens a window (Figure 6P) which tells you what version of the System and Finder you're using, how much memory your computer has, and how much memory is available for running applications. The amount of memory used by the Finder, the System, desk accessories, and setup files is also displayed. This window is a standard IIGS window and can be dragged, moved behind other windows, and closed.

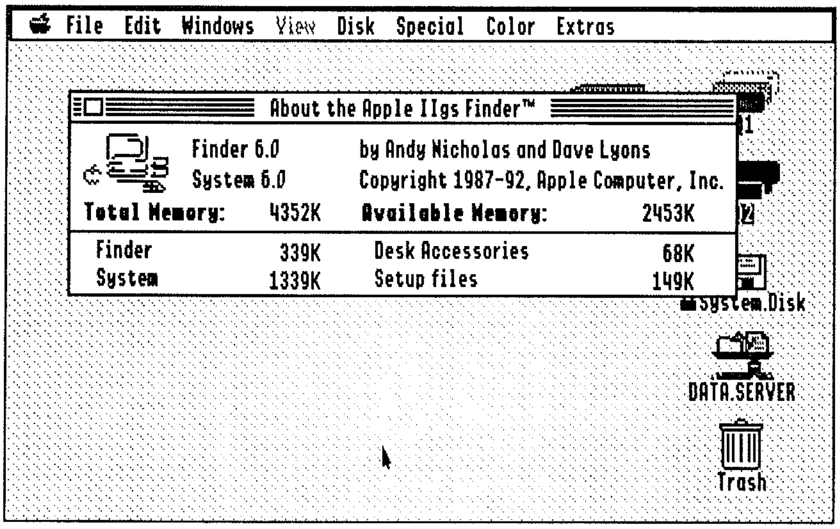


Figure 6P—About The Finder

Help (Menu)

Choosing "Help" displays the Finder Help window. At the top of the window is a pop-up menu which allows you to select the topic you need help on. All the basics of Finder operation are right there, including tunneling, file copying, selecting and coloring icons, and shortcuts. Like the "About" window, this window can be moved around. You can scroll it to view the contents of long help sections, and you can leave it open while you're using the Finder to have easy access to the help any time.

Validate (File Menu)

We touched on this feature briefly when we talked about verifying disks. Individual files can be validated, as well. If you select more than one icon, they will be validated separately, with separate summary windows. For this reason, if you want to validate all the items in a particular folder or disk, it's best to select the disk or folder instead of the files themselves. Validating a disk or a folder validates all the files in the folder, including the contents of any folders, *ad infinitum.* If any unreadable files or folders are found, a summary window can be displayed; the summary window is a standard Finder window and can be dragged, closed, and can remain open while you continue to use the Finder.

Show Clipboard (Edit Menu)

The Show Clipboard menu item displays the contents of the IIGS Clipboard, which can contain text, graphics, or other data. You can use this feature to verify that the data you cut from one application is still on the Clipboard before launching another application. The clipboard window can be repositioned, resized, and scrolled, and can stay open while you use the Finder.

Preferences (Special Menu)

The Finder Preferences Dialog (Figure 6Q) lets you configure certain aspects of the Finder's operation, so you can customize the Finder to the way you work. (Like most good programs, the Finder is flexible enough to accommodate nearly everyone's needs.)

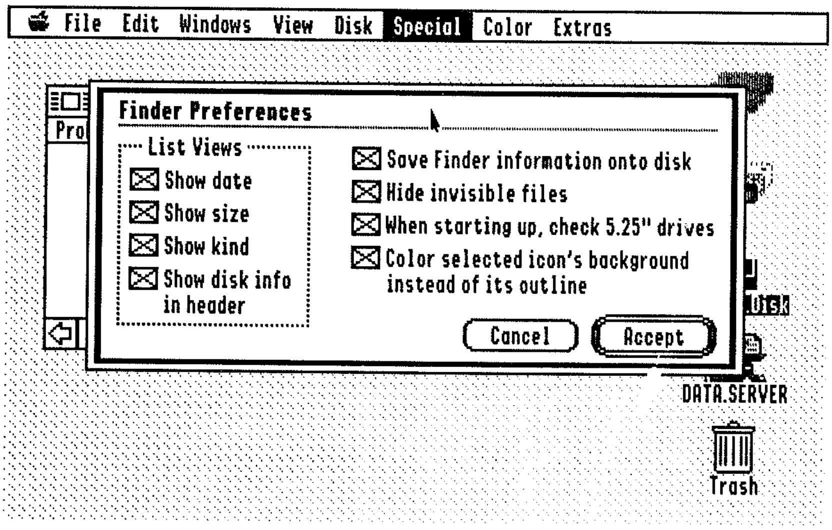


Figure 6Q—Finder Preferences

The four checkboxes listed under "List Views" allow you to select which columns of information you want to see in the list views ("By Name," "By Date," "By Size," and "By Kind"). If you don't often care when a file was last modified, for instance, you can turn that column off by clicking the "Show date" checkbox.

The "Show disk info in header" also affects list views. Usually list views have *two* lines of header information. The first line tells you the disk's type (ProDOS, HFS, etc.), the number of files in the window, and how much space is used and available on the disk. The second line provides labels for the columns. If you'd rather see more files and less disk information, turn off this checkbox and the first line will disappear. (Icon views always show the disk info.)

"Save Finder information onto disk" controls whether or not the Finder saves the locations, sizes, and views of your windows and the positions and colors of your icons. The Finder saves this information in hidden files known as Finder.Data files—one in every window you've ever opened in the Finder. If you'd rather not have the Finder leaving its droppings all over your disks, turn this checkbox off. Any Finder.Data files the Finder has already saved will remain.

☞ **Note!**

You can override this feature temporarily by holding down the Control key while you close a window. The Finder won't save the data for that window. (If the Finder has previously saved the information in this folder, it will not be updated—the next time you open the folder, any changes you made to the appearance of the window will be gone.) The Control key forces the Finder to do the opposite of the Preferences setting—if you turn "Save Finder info" off, holding down the Control key when you close a window will cause it to be saved.

▲ **By The Way...**

If you hold down Control while you open a folder or disk, the Finder will ignore any information it's already saved for that window and display it at the default location with the default size (the view will be set to "By Icon.")

If you'd like to be able to see Finder information files and other normally hidden files, turn the "Hide invisible files" checkbox off. (Turn off "Save Finder info" and "Hide invisible files," to seek out and destroy Finder info files without recreating them as quickly as you delete them.)

▲ **By The Way...**

The Finder saves information in the following files: Finder.Data (position of windows and icons), Finder.Root (position of icons moved to the desktop), and Finder.Def (preferences).

If you uncheck "When starting up, check 5.25" drives," the Finder will display only the 5.25" drive icon, and won't try to read the disk. You'll have to double-click the drive icon to make the Finder look in the drive. If you have 5.25" drives but rarely use them, you can uncheck this box to avoid the nasty grinding sound the Finder makes when it checks empty drives.

If you uncheck "Color selected icon's background instead of its outline," the Finder's Color menu will color an icon's outline instead of filling it with color. (You normally get this effect by holding down the Control key when you choose a color; changing this preference *reverses* the function of the Control key when choosing colors.)

Once you've selected your preferences, click "Accept" to save them. Your new preferences take effect immediately. If you didn't really want to change the preferences, click "Cancel."

Icon Info (Special Menu)

Selecting an icon and choosing Icon Info (or pressing ⌘-I) will display an "Info On..." window (Figure 6R). You can get info on files, folders and disks—even the Trash, if you feel so inclined. The info window can be moved around and behind other windows—it's a standard Desktop window. In the window you'll see up to four "index cards" (some icons will have fewer cards). Click the tab on the desired index card to flip to that card. The four cards are:

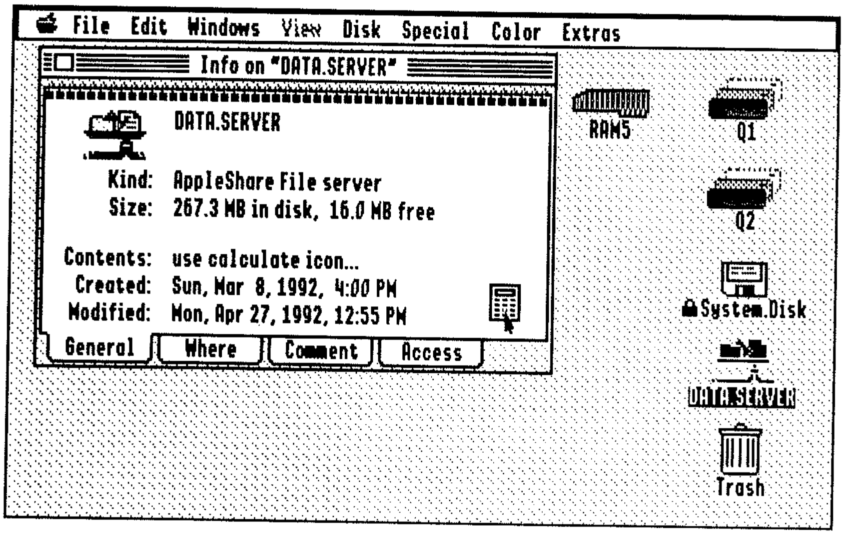


Figure 6R—Icon Info Window

- *General:* Tells you what the icon represents (e.g., "ProDOS 3.5" Disk"), how much of the disk's storage space is used (or how much is used by a particular file), and when it was created and last modified. There's a calculator icon on disk and folder cards which, when clicked, causes the Finder to scan through the directory and report exactly how many files and folders are contained by the icon. For file icons, there's a checkbox to lock the file (protect it from accidental deletion or modification) and, on many system icons (desk accessories, drivers, and so forth) there's a checkbox to deactivate the file to keep the system from using it on the next startup.

- *Where:* Tells you the name of the device the icon is on (e.g., APPLEDISK3.5A) and the icon's complete pathname. The icon's pathname lists all the folders it's in (starting with the disk it's on, working its way down through the folder hierarchy, and ending with the name of the icon itself).

- *Comment:* Allows you to add comments to your files and edit these comments. If the file doesn't have a comment, an "Add Comment" button will appear; click it to add a comment.

Warning!

Once you've added a comment to a file, you can't access the file from older (ProDOS 8) programs. For this reason, you probably won't want to add comments to, for example, your AppleWorks files—AppleWorks would no longer be able to read them! IIGS Desktop programs would, however, still be able to access the files. An exception to the rule—if you move the file to a network volume, you'll be able to access it again. You'll get an alert to this affect when you add a comment to a file; click "Attach" if you decide to really attach a file.

- *Access:* Displays your access privileges on shared disks (network volumes). Access privileges can be changed in the FolderPriv control panel.

Shortcut

You can use the keys ♁-1 through ♁-4 to view the four cards of the Info window.

Shut Down (Special Menu)

Use the Shut Down option at the end of a Finder session. When you select "Shut Down" from the Special menu, the Finder Shut Down Options dialog (Figure 6T) appears. Your options are:

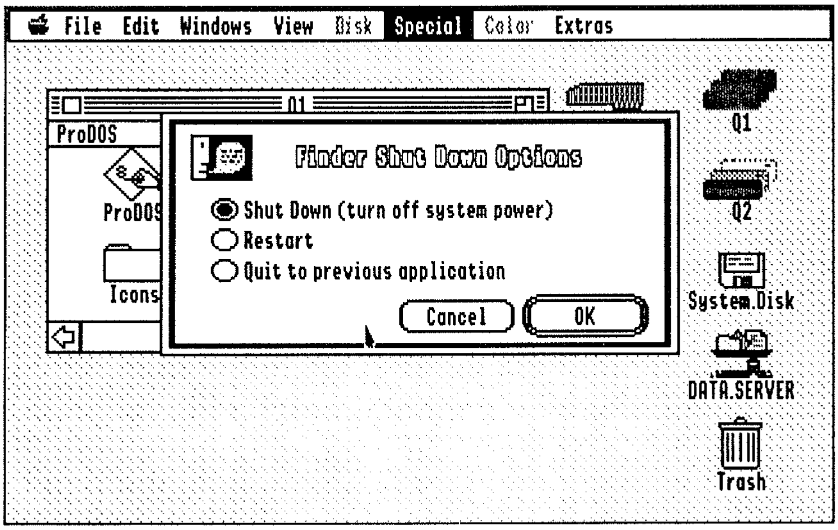


Figure 6T—Shut Down Dialog

- *Shut Down (turn off system power):* Use this option at the end of a computing session to make sure all files are saved. The Finder empties the trash, saves the position of all open windows (if the "Save Finder info" preference is on), ejects all 3.5" disks, and presents an alert that tells you it's safe to turn off the computer. It's not always necessary to use this option—in fact, you can turn the computer off in the middle of an application program if you want—but returning to the Finder and shutting down insures that you remembered to save all your files and that you didn't leave any disks in your drives. You neat freaks will always shut down this way.

- *Restart:* Performs a shutdown, then automatically restarts the computer. Do this after installing new system elements (desk accessories, fonts, etc.).

- *Quit to previous application:* The Finder is a Desktop application and it's the one most IIGS users start up into, but it's possible to start up into a different application (for example, a program launcher such as Wings or ProSel) and only run the Finder when necessary. To return to your original launcher, choose this option.

☞ **Note!**

If you choose "Quit to previous application" but there is no previous application, System 6's built-in "mini-launcher" (described in the next chapter) will appear.

Once you've selected your shutdown option, click "OK" to shut down, or "Cancel" to return to the Finder without shutting down.

➠ **Shortcut**

You can press the S, R, and Q keys to choose "Shut Down," "Restart," or "Quit." (You can also use the up and down arrow keys to move the radio button from one option to another.)

The Extras Menu

Since the Finder can't be all things to all people, Apple provided a way for programmers to add new features to the Finder via Finder extensions. These new features appear in the Extras menu (for the most part—some extensions don't have a menu item but instead enhance some other Finder feature). If you haven't installed any Finder extensions, you won't have an Extras menu.

▲ **By The Way...**

Finder extensions are installed by copying them into the System.Setup folder (inside your System folder) on your hard drive or startup disk. Alternately, yo'. can create a folder inside the System folder called FinderExtras and copy them into that folder. Finder extensions in System.Setup are loaded at startup time and take up memory no matter what program you're using. Extensions in FinderExtras are loaded only when the Finder is active; they don't take up any memory when you're using other programs, but they must be reloaded from disk whenever you return to the Finder, which can slow things down.

The Quality Computers System 6 Bonus Pack includes two Finder Extensions, called IR and QuickLaunch. These Finder Extensions are covered in a later chapter.

Chapter Seven

WORKING WITH APPLICATIONS

7

More About The Desktop

As we've been mentioning pretty much continuously throughout this book, most Desktop programs operate in pretty much the same way. In Chapter 5 we covered the features that most Desktop programs have in common. In Chapter 6, we took an in-depth look at the Finder and covered concepts pertaining to documents, files, and folders at the same time.

Since you know about using the Desktop's elements and how to launch applications from the Finder, you know enough to use most Desktop-based programs. However, you'll get further faster if we give you a few more pointers on dialogs you'll see again and again as you use Desktop applications under System 6.

One of these dialogs is the Standard File dialog. You use a Standard File dialog when you tell an application which document you want to work with and when you save and name a document. System 6 provides Standard File dialogs to keep the user interface consistent—it's just one more way to make all programs work the same.

You'll also see standard dialogs when you print your documents. One dialog, Page Setup, allows you to tell the application what size paper you'll be using and other information the computer needs to determine how to divide your document into pages. The Print dialog appears when you actually print your document and allows you to specify the number of copies you want printed, which pages to print, and whether to use draft or letter quality print.

The other dialog we'll cover is the mini-launcher. We mentioned it in passing in the last chapter; it's a small program that appears when you quit a program and there's no previous application for the program to quit to. (In reality, it's little more than a Standard File dialog.)

Remember, these features are a part of System 6 itself and are available in all Desktop programs. Learning how to use these dialogs will make Desktop programs even easier to use.

Let's get into a program that works with documents so you can follow along in our discussion. We suggest the Teach program from the "SystemTools2" disk. (It may also be on your hard drive if you installed System 6 there.) Teach is a sort of miniature word processor that's intended for reading and creating short text documents. The Finder uses it as the default application for text files. For now, just double-click the "Teach" icon to start the program.

The Standard Open Dialog

Pull down Teach's File menu and select "Open" to see an Open dialog. If you're running the program from the SystemTools2 disk, the dialog will look like the one in Figure 7A.

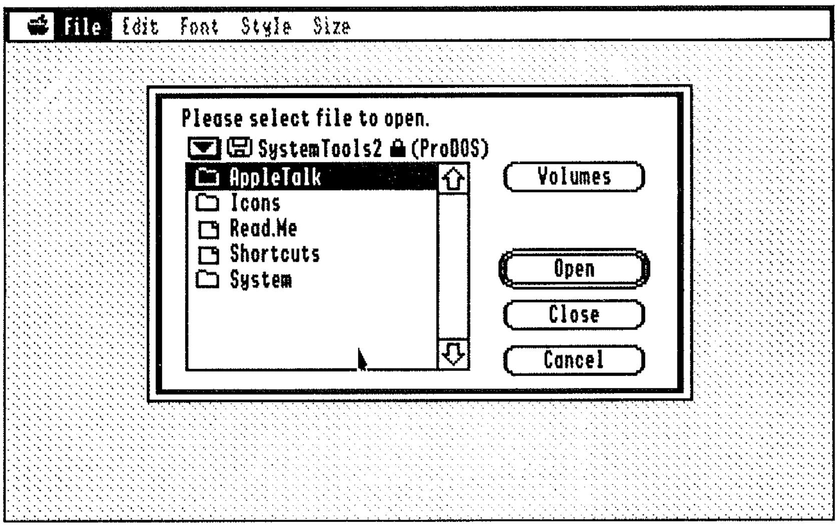


Figure 7A—Standard Open Dialog

At the top of the dialog, the program tells you what you're supposed to do—in this case, "Please select file to open." Each application can fill in this message—some are less formal. (We know of at least one program that simply asks "Whaddaya want?") Below this instructional message is an icon representing the disk you're using, the name of the disk, and its type: ProDOS, in our case. (There's also a padlock icon there if your disk is write-protected.)

The dialog is dominated by an alphabetical list of the files on the disk. The list will have a scroll bar if there are more files than will fit in the window. In the example, we're looking at the main SystemTools2 directory, which contains three folders and two documents.

☞ **Note!**

Most Open dialogs show only the files that the program can open, along with folders. If you look at SystemTools2 with the Finder, you'll see that there are two applications (Teach and Archiver) in the main directory in addition to the documents. Teach doesn't show you those files in its Open dialog because Teach doesn't open applications; it opens text documents. The same thing would happen if there were graphics documents or other non-text documents on the disk; they, too, would not appear in Teach's Open dialog.

▲ By The Way...

Some applications show all the files in a directory, but the ones the program can't open are dimmed so you can't select them. You can only select files that are displayed in black.

Let's open the "Read.Me" file. This document tells you who in the System 6 development team did what. First, click the Read.Me file; then click the Open button. The file will be opened and displayed on your screen, and Teach will be ready for you to read or edit it.

The Open button has a double border, making it the default button, so we could have simply pressed Return to open the file once we'd selected it. (Most applications also accept ⌘-O to trigger the Open button.) Or we could simply have double-clicked the file we wanted to open. There's a Cancel button there, too, which obviously would dismiss the Open dialog without opening any file (you can also use its key equivalent, ⌘-Period.)

Instead of clicking the file we wanted to open, we could have used the up and down arrow keys to select it. We could even have typed the first letter of the file's name, and the black bar would have jumped immediately to the first file in the list that starts with that name. In fact, you can type the first *several* characters of the file's name (as long as you type them quickly enough) and the black bar will jump to the first file starting with the letters you type.

For now, make no changes to the "Read.Me" file. Just close it. (You can click the close box in the upper left corner of the window, or use the "Close" option on the File menu or its key equivalent.) Now bring up the Open dialog again—we'll take a look at some of the other features of the Standard File dialog.

Let's look inside a folder. Double-click the System folder. Not surprisingly, there are no files in the System folder that Teach knows how to open, so you just see a lot of folders. Double-click the Sounds folder to look in there. Nope, no openable files there. Click the Close button (or press Escape) to move back upward through the hierarchy to the System folder, then double-click the Fonts folder. Disappointment—no openable files there either. (There aren't any files on the SystemTools2 disk that Teach can open other than the two in the main directory.)

Press the downward-pointing triangle next to the disk name to access the pop-up folder menu. This menu shows you how deep in a disk you are, starting with the current directory and working backward through parent directories to the disk name. In our case, we see the Fonts folder, the System folder, the disk's main directory, and, curiously, a level called "Desktop."

From this pop-up menu, you can select any directory in the hierarchy and the Standard File dialog will immediately display the contents of the directory For example, we could select "SystemTools2" and immediately "pop back" to that directory. It should be becoming clear that the Standard File dialog represents your disk's directories in a sort of "flattened" fashion. (In Standard File, unlike the Finder, you see only one directory at a time.)

▲ By The Way...

If it helps, think of the Standard File directory as having built-in tunneling. When you open a new folder, the old one is automatically closed.

Selecting the "Desktop" level from the pop-up folder menu or clicking the "Volumes" button displays the names of all the disks in drives connected to your computer (Figure 7B). Double-click on the disk you want to use and you'll immediately move to that disk's main directory. The "Desktop" or "Volumes" level lets you switch to any other disk you want to use in seconds.

Note!

In some Standard File dialogs, the "Volumes" button may read "Disk" or "Drive." It serves the same function regardless of its name.

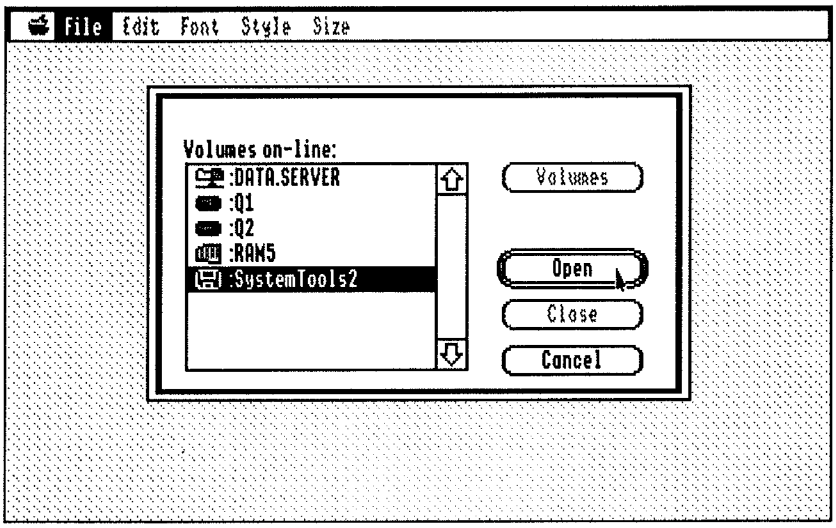


Figure 7B—Volume Selection (in Standard Open)

Some applications add other buttons to the Standard File dialog. These buttons' functions differ from program to program. Consider Teach's Import File dialog—note the four radio buttons at the bottom of the dialog that tell Teach what kinds of files you want to see in the dialog. But the basic elements of the Standard File dialog are still there at the foundation.

Shortcut

The following shortcuts are available in most Standard File dialogs:

⌘-Tab	Move to next disk (like going to Volumes and selecting it)
⌘-Period	Cancel (same as clicking Cancel button)
⌘-Up Arrow, Escape	Move to parent folder (same as Close button)
⌘-Down Arrow, ⌘-O	Open folder or document (same as Open button)
⌘-Escape, ⌘-D	Jumps to the Volume list (Disks)
Insert 3.5" disk	Display that disk's main directory

The Multi-File Open Dialog

A multi-file Open Dialog looks much like a Standard Open dialog, except that in addition to having a button marked "Open," it has one marked "Accept." And the message at the top of the dialog will tell you to select the *files* you want to open, not the (singular) file. You can select multiple files by holding down the ♂ key while clicking second and subsequent files in the list. (You can also select a range of files by clicking the first file in the group and holding down Shift while you select the last file in the group.) To unselect a file you selected in error, hold down ♂ and click the file.

When you've selected the file or files you want to open (you can select just one if you like), click the "Accept" button.

The Standard Save Dialog

The Standard Open dialog is used when you tell an application you want to work with an existing file—when opening files, or even, in some cases, when deleting a file or performing other operations on it. The Standard Save dialog (Figure 7C) appears when you save a document you haven't named yet (using the "Save" option on the application's File menu) or when you save an existing document under a new name (using "Save As").

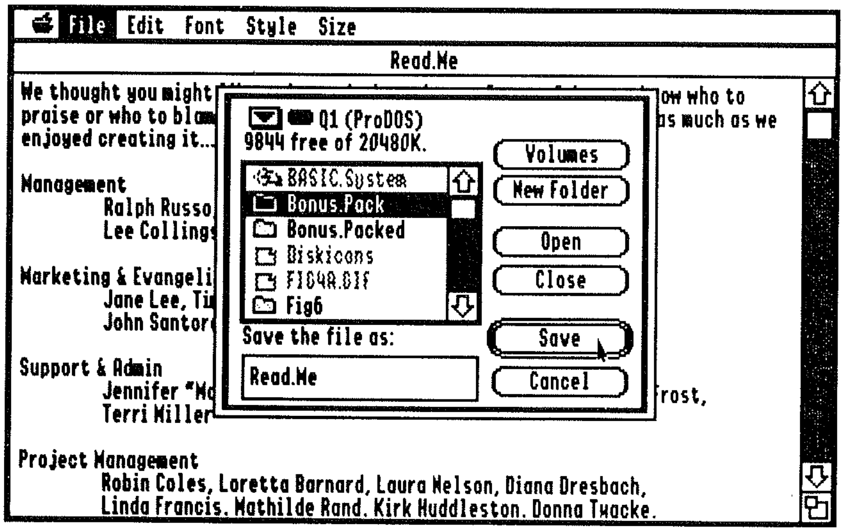


Figure 7C—Standard Save Dialog

The Standard Save dialog has all of the features of the Standard Open dialog, plus a "New Folder" button and a line-edit field for entering the name of your document. You may also notice that the Save dialog shows *all* the files in the current directory (dimmed, of course), not just the ones the application can use, so you can make sure that the name you give your document isn't already used by another file.

To save your document in the current directory, just type its name (or edit the existing name) and press Return. To move to another directory first, you can scroll through the list and open folders just as in the Open dialog; the pop-up menu and Volumes button work as usual too. (To type a file or folder's name to jump to it immediately, you must first press Tab to make sure the file list has a bold black border around it. Otherwise, whatever you type will go into the dialog's line-edit field.)

To save the document into a new folder, first type the name of the folder into the line-edit field, then click the "New Folder" button. A new folder will be created and opened inside the current disk or folder. Now type the name of the document and click "Save" (or press Return) to save the document in the new folder. If you give the document a name that matches an existing file, you'll get an alert that asks you if you really meant to do that—click "Replace" in that dialog to save the new document and remove the old one.

☞ **Note!**

The "Save" button has a double-border in the Save dialog and will be activated by pressing Return. To open an existing folder, use the "Open" button (or its keyboard equivalents)—you can't just press Return as you can in the Open dialog.

➠ **Shortcut**

The shortcuts in the section on Standard Open dialogs also work with Standard Save dialogs.

More About Documents

Documents are at the heart of everything you do on the IIGS. Documents reside on disk while you're not working on them; they're copied into the computer's memory when you open them. The file on the disk isn't updated unless and until the document is saved.

The File menu in most applications has several options for managing your documents.

- *New:* Begins a new document. The new document is usually called "Untitled" until you save it for the first time, at which time you give it a name.

- *Open:* Starts working with an existing document. Many applications allow you to work with more than one document at a time. Those that don't will remind you to save your current document before you open a new one.

- *Save:* Saves the current document to disk. If the document has never been saved, a Standard Save dialog appears to allow you to name the document and specify the directory in which to save it. If the document has already been named, the document is saved back to disk in its original location without presenting any dialogs or alerts, updating your disk copy of the document to be the same as the document in memory.

- *Save As:* Allows you to save an old document under a new name by presenting a Standard Save dialog. When you do a Save As, the original file remains untouched and you begin working with the new copy of the file—all future Save operations will update the new copy without changing the old one.

- *Close:* Closes the document you're working on. If you've made changes to the document but haven't saved them yet, the application asks if you want to save the changes and performs a Save operation if so.
- *Revert:* Throws away the changes you've made to the document since the last Save operation and opens a fresh copy of the document from the disk.

▲ By The Way...

Some programs call Standard File dialogs by the names programmers use when writing software. (You'll usually see these terms when the same person wrote both the program and the manual). For the record:

Programmerese	English
SFGetFile	Standard Open Dialog
SFMultiGetFile	Standard Multi-file Open Dialog
SFPutFile	Standard Save Dialog

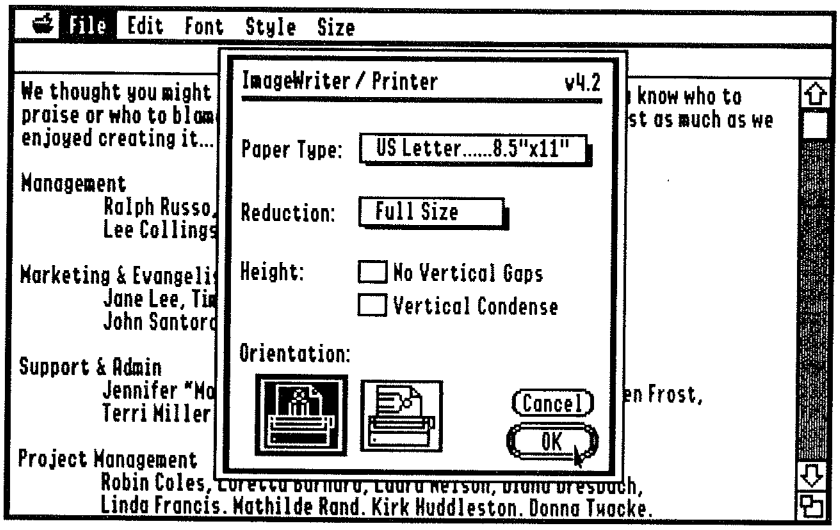


Figure 7D—Page Setup Dialog (ImageWriter)

Printing Documents

The Page Setup Dialog

When you create a document, most applications assume that you'll be printing upright on a standard letter-size sheet of paper. If you won't, you can use the File menu's "Page Setup" option to tell the application what size paper you *will* be using, and set other printer options such as reduction and enlargement, condensed printing, and so on.

It's important to adjust your page setup as soon as possible so that the application can accurately display your document on the screen. (This feature, called What You See Is What You Get, or WYSIWYG, allows you to lay out your document on the screen and be sure that your final printout will look pretty much the same.)

The actual appearance of the Page Setup dialog depends on which printer you've selected in the Control Panel, since each printer supports different paper sizes and resolutions. We'll cover the ImageWriter Page Setup dialog (Figure 7D) in detail, then pointing out the parts of other such dialogs that differ from the ImageWriter dialog.

At the top of the Page Setup dialog you'll always see your printer's name and the port it's connected to. In this case, we've got an ImageWriter connected to the Printer port. (Other choices for the printer include StyleWriter, ImageWriter LQ, and Epson; other choices for the port include Modem port, Parallel Card, or AppleTalk. The LaserWriter Page Setup dialog simply says "LaserWriter.") If this information is not correct, use the DC Printer or Net Printer Control Panels to choose a printer. (More information on these Control Panels appears in Chapter 9.)

Next, the ImageWriter Page Setup dialog has a pop-up menu to select the paper size. The ImageWriter supports standard U.S. and international paper sizes, as well as #6 envelopes and a variety of labels. (Other printers may support fewer sizes and use radio buttons rather than a pop-up menu to choose among them.)

Below that is a pop-up menu to select the document's intended print size—either full-sized, or reduced to 50%. Some printers have more choices here—the ImageWriter LQ lets you select 33% or 66% sizes (but not 50%) and the LaserWriter has a line-edit field that lets you select any percentage reduction (less than 100%) or enlargement (greater than 100%). The Epson driver has a checkbox for 50% adjustment.

The two checkboxes below allow you to select No Vertical Gaps (blank space at the top or bottom of pages) and Vertical Condense. The IIGS screen doesn't have a square aspect ratio—that is, if you draw a square on the screen with exactly the same number of dots both horizontally and vertically, it will look like a tall rectangle because the dots aren't square.

If Vertical Condense is turned off, what appears on your printer will look just like what's on your screen—if you draw a circle that looks right on your screen, it will look right on the printer. However, most text will look too tall. If you turn on Vertical Condense, text will be the right size but graphics may look squashed.

The Epson driver displays radio buttons for selection of Normal or Condensed Vertical Sizing. These are the same as Vertical Condense off and on, respectively. The LaserWriter allows you to select Normal (same as Vertical Condense off), Intermediate (a compromise setting that squashes graphics a little and stretches text a little), and Condensed (same as Vertical Condense on). The StyleWriter supports Normal (Graphics) and Condensed (Text) settings.

Finally, in the ImageWriter Page Setup dialog, you can select upright (portrait) or sideways (landscape) printing modes. Most other printers (except for the StyleWriter) also print sideways or upright.

The ImageWriter LQ has two additional checkboxes for darker printing and unidirectional printing. With unidirectional printing, the print head only prints when traveling from left to right, for better alignment in graphics.

The LaserWriter has two checkboxes for smoothing and font substitution. Smoothing can help reduce the jagged edges encountered when printing graphics and certain kinds of text. Font Substitution makes the printer use Helvetica in place of Geneva and Courier in place of Monaco when printing text; since Helvetica and Courier are built into the printer but Geneva and Monaco are not, this can improve the print speed and the quality of the final output.

The StyleWriter driver has a pop-up menu to allow you to choose either automatic or manual sheet feed and another to select one of two print qualities. With manual sheet feeding, the computer will pause and ask you to insert a sheet of paper before each page is printed. Changing the print quality allows you to trade off your time for better print quality—360 DPI printing is much slower than 180 DPI.

Once you've set your Page Setup options, click "OK" or press Return to accept them. Page Setup options are saved with each document. If you decide you don't want to change the options after all, click "Cancel" or press Escape. Now you're ready to work on your document.

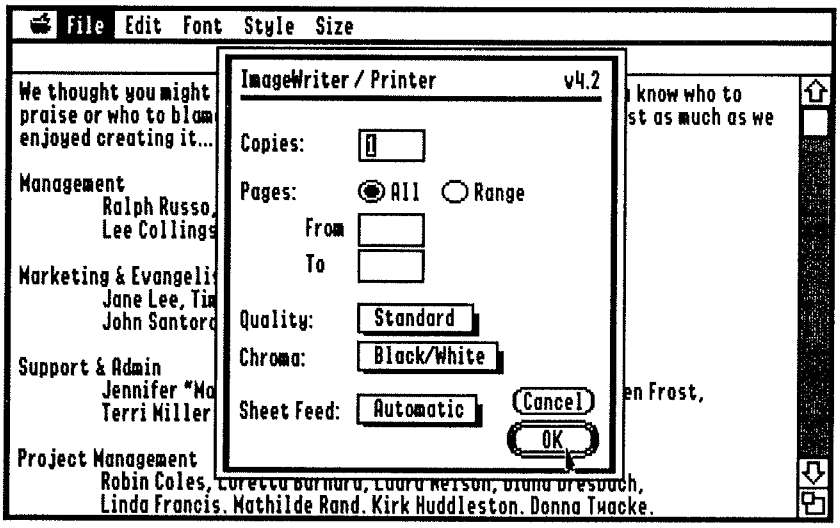


Figure 7E—Print Dialog (ImageWriter)

The Print Dialog

Once you've completed a document and are ready to print it, choose "Print" from the File menu. The Print dialog (Figure 7E) will appear. As before, we'll concentrate on the ImageWriter version of this dialog and mention how other printers' Print dialogs differ from it.

The first thing is a line-edit field for specifying the number of copies to be printed. Below that are radio buttons that allow you to print all of the document or only a range of pages and two line-edit fields for specifying the starting and ending pages. As usual, you can use the Tab key to move from one line-edit field to another. (Tabbing into the From or To fields automatically switches the "Pages" radio buttons to "Range"—thus, to print a range of pages, just hit Tab after specifying the number of copies and enter the starting page number, then tab again and enter the ending page number.)

The ImageWriter driver supports four print modes, which can be selected from the Quality pop-up menu. Text-only uses the built-in ImageWriter typeface to print a draft of your document without any graphics; Fast prints a draft of your document using the typefaces you chose and the graphics you included. Standard does a little better job, and Best yields the best results but takes a long time to print and uses more ink. The Epson driver supports three print modes—Better Text, Better Color, and Draft.

The Chroma pop-up menu allows you to choose either black and white or color printing (the option for color printing is dimmed if a color ribbon isn't installed in the printing). And the Sheet Feed option lets you select whether the printer stops before every page to allow you to insert a sheet of pre-cut paper. The StyleWriter driver also allows you to re-specify the print quality (360 DPI or 180 DPI) in the Print dialog.

Once you've chosen your options, click "OK" or press Return and the printing operation will begin. (Or click "Cancel" or press Escape to exit the dialog without printing anything.)

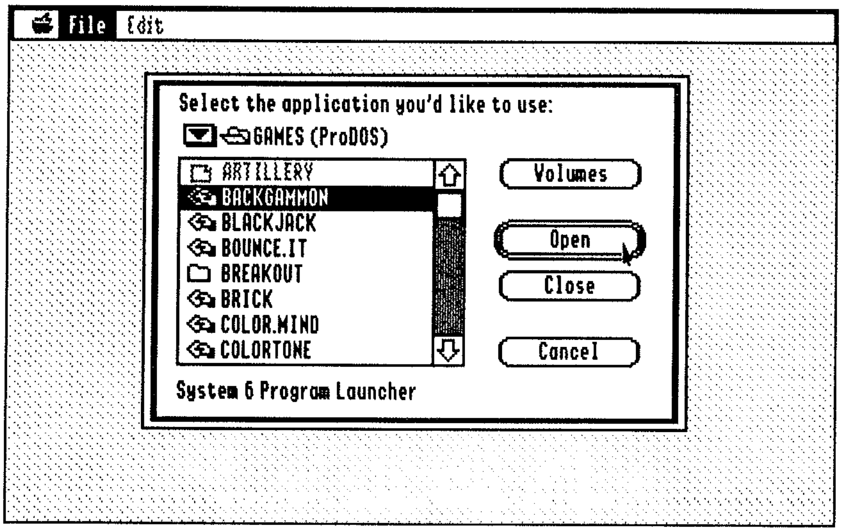


Figure 7F—The System 6 Launcher

The System 6 Launcher

On occasion, when quitting a program, you'll see the System 6 Launcher (Figure 7F). This launcher appears when you quit a program and there's no "previous" application to return to (for example, when quitting the Finder), or when you boot System 6 and the application you set in the SetStart Control Panel is not available.

The Launcher is a Standard Open dialog that allows you to select the next application you want to use. Both Desktop and non-Desktop programs are available, though BASIC applications are not. Find the program you want to launch as usual, using the "Volumes" button and the pop-up folder menu and by opening the appropriate folders to get to the application. Then double-click the application to launch it.

If you want to shut down the computer, click the "Cancel" button, then select "Shut Down" from the File menu.

☞ **Note!**

The Finder is a standard application file and can be launched by the Launcher. Just open the System folder of your startup disk and launch the file called Finder.

Chapter Eight

WORKING WITH DESK ACCESSORIES

About Desk Accessories

The Apple IIGS supports two different kinds of desk accessories. Desk accessories are the electronic equivalent of the tools you keep on your real desk—things like calculators, notepads, and calendars—that are usually kept to the side of your regular work but are always available at a moment's notice. (To the computer, of course, desk accessories are just more programs.)

Classic Desk Accessories, also called CDAs or text desk accessories, use the IIGS's text screen (the screen used by AppleWorks and most older programs). Most are accessible from both Desktop and non-Desktop programs. To see a list of available CDAs, hold down and Control while pressing and releasing Escape.

New Desk Accessories, also called NDAs or graphic desk accessories, appear in windows in your Desktop programs. NDAs are listed in your menu.

Desk accessories (both NDAs and CDAs) are installed by placing them in the Desk.Accs folder inside your startup disk's System folder. After you install any desk accessories, you must restart the computer before they are available for your use.

▲ **By The Way...**

If you bought the Quality Computers System 6 Bonus Pack, there's a Finder Extension called IR that allows you to install desk accessories without having to restart, and without the desk accessories needing to be in the Desk.Accs folder. Keep reading for more details.

Finder extensions are similar to NDAs.—they're essentially desk accessories that work only in the Finder. Instead of appearing in your menu, Finder Extensions appear in the Finder's Extras menu (which appears only if you have installed one or more Finder Extensions). Finder extensions are installed by placing them in the System.Setup folder inside your startup disk's System folder.

▲ **By The Way...**

Finder extensions can also be placed in a folder called FinderExtras—this folder doesn't exist on a normal System disk, so you must create one first. Finder extensions installed in the FinderExtras folder only use memory while you're in the Finder but cause the Finder to take longer to start up; extensions in the System.Setup folder use memory all the time but don't slow down the Finder.

In this chapter, we'll discuss the CDA menu (and the enhancements made to it under System 6) briefly. There are no new CDAs included with System 6, though (there's one in the System 6 Bonus Pack), and the old ones work the same way they always have. The text-based Control Panel, for example, is a built-in CDA and it's well documented in your IIGS Owner's Manual, so we won't cover it in this manual.

We'll be covering the NDAs that come with System 6 in some depth, though, along with the Finder extensions. These accessories are all new or completely revised in System 6—plus there are even more in the System 6 Bonus Pack! (Bonus Pack desk accessories are marked with an asterisk. If you didn't buy the Bonus Pack, you won't have these accessories.)

Classic Desk Accessories

Classic Desk Accessories (CDAs) are accessed by holding down ⌘ and Control while pressing and releasing Escape. You select the CDA you want from the list displayed on the screen (Figure 8A) by using the Up and Down arrow keys; press Return to open the highlighted CDA.

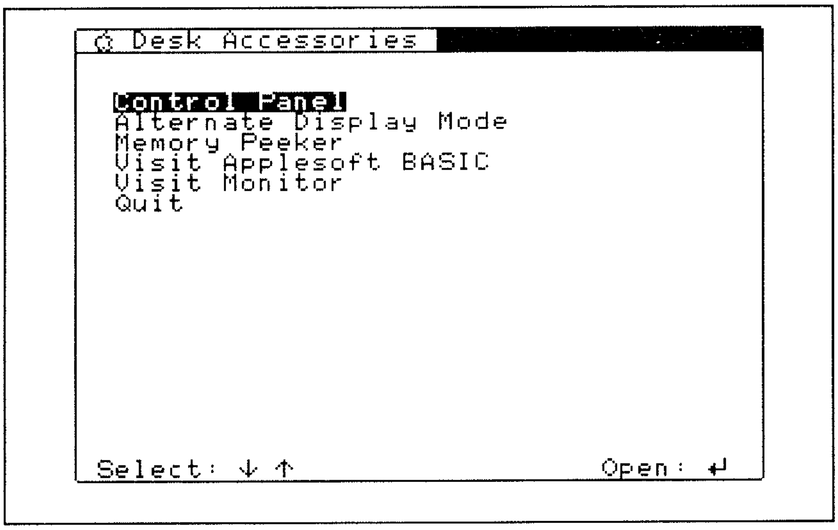


Figure 8A—Classic Desk Accessory Menu

⑥ New & Improved

System 6 enhances the CDA menu to allow you to use ⌘-Up Arrow and ⌘-Down Arrow to move the highlight bar a screenful of CDAs at a time (if you have that many installed). You can also move the highlight bar to the first CDA starting with a certain letter by pressing that letter. Pressing Escape moves the highlight bar directly to "Quit." Some of these features were previously available on the ROM 03 IIGS; System 6 brings them to all IIGS models.

Activating The Built-In CDAs

All Apple IIGS models have two built-in CDAs called "Visit Monitor" and "Memory Peeker." Depending on which Apple IIGS model you have, though, these CDAs may not appear in the CDA menu automatically. While documenting the use of these CDAs is beyond the scope of this manual (they're rather technical and useful mainly for debugging programs) we're not beyond telling you how to install them:

- Double-click the BASIC.System icon on your startup disk
- When the BASIC prompt (]) appears, type "CALL-151" (without the quotes) and press the Return key.
- When the Monitor prompt (*) appears, type a number sign (#) and press Return.
- Finally, type "BYE" (without the quotes) and press Return.

Upon returning to the Finder, access the CDA menu and note the two new CDAs. The CDAs will disappear each time you turn off the computer.

*Visit Applesoft BASIC**

The System 6 Bonus Pack includes a new CDA called "Visit Applesoft BASIC." If you're familiar with the IIGS's built-in "Visit Monitor" CDA, you'll like this one. From any Desktop program, select "Visit Applesoft BASIC" from the CDA menu and you're plopped into Applesoft BASIC.

Disk commands are not available, and you should not use graphics commands if you want to avoid crashing your computer. Press an ampersand (&) followed by Return to exit to the CDA menu. This CDA does not work from ProDOS 8 programs.

It's mainly for hackers—for people who occasionally would like to pop into BASIC and do a quick calculation or cobble together a short program to test some idea they've had. But we thought some of you would find it useful, so we included it.

New Desk Accessories

New Desk Accessories (NDAs) are activated by choosing them from the menu of most Desktop applications. NDAs appear in their own windows which can be manipulated like any other Desktop window. You can even leave NDAs open while you work on something in your main application program—a click in their window will bring them back to the front. You can have a virtually unlimited number of NDAs open at any time.

NDAs can be closed by clicking their close box (in the upper left corner of their window) or by choosing "Close" from the File menu or by pressing ⌘-W. Additionally, all open NDAs are closed automatically when you quit a program.

Calculator

The calculator NDA (Figure 8B) is a simple four-banger that can be operated with the mouse or via the IIGS numeric keypad (notice that the calculator's on-screen layout is the same as the numeric keypad's physical layout; the asterisk, "*" is used for multiplication and the slash, "/" is used for division). The calculator doesn't understand precedence—it performs all operations in the order you choose them. (If you're used to algebraic calculators where multiplication and division are always performed before addition and subtraction, well, this one isn't quite that sophisticated. The System 6 Bonus pack includes a scientific calculator that's quite a bit more powerful, though.)

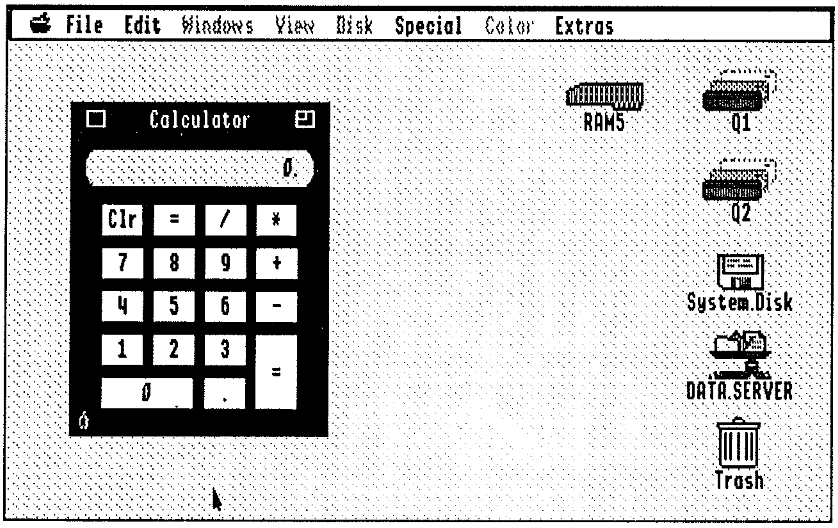


Figure 8B—The System 6 Calculator

▲ By The Way...

Clicking the Calculator's zoom box (upper right corner of its window) or pressing the 'Z' key expands the calculator's keypad to include a hexadecimal keypad. Click the "Hex" button to switch the calculator to hexadecimal mode. (You programmers will know what that is—it's next to useless for ordinary mortals, though.) Pressing the 'H' key toggles the calculator between hexadecimal and decimal mode. The calculator's hexadecimal keypad does not need to be visible to be used.

Shortcut

Use the "=" key to repeat the last operation. For example, to count upward by odd numbers, type "1 + 2 = = = = =." Typing "/" followed by "=" calculates the reciprocal of the number in the display (same as dividing it into 1). For example, "4 / =" yields 0.25.

The contents of the calculator's display can, of course, be cut and pasted. "Copy" copies the contents of the calculator's display to the clipboard, ready for pasting into a spreadsheet or other document. "Cut" does the same *and* clears the calculator's display. "Paste" passes the contents of the clipboard to the calculator just as if you were typing it on the keyboard. This means you can paste an entire expression (like 5 / 7 + 25 =) to the calculator and get the result in the display. Any keystrokes that the calculator doesn't recognize will be ignored.

CD Remote, Media Controller, & VideoMix

These NDAs are part of Apple's Media Control package. CD Remote allows you to play audio CDs on an Apple CD-ROM drive using an on-screen "remote control." Media Controller lets you control videodiscs and other devices, also using an on-screen remote. VideoMix lets you control the Apple Video Overlay Card from within any Desktop application.

Control Panels

The Control Panels NDA is used to access the System 6 control panels. Control panels are small accessory programs, much like desk accessories, which are used to configure the operation of your IIGS, System 6, and other things.

There are so many control panels that we've given them a chapter of their own (the next one, in fact). We'll skip over this NDA for now.

CloseView & Video Keyboard

Video Keyboard displays a windoid with a simulated Apple IIGS keyboard, which you can click instead of typing. CloseView allows you to enlarge the screen display. These two NDAs are part of Apple's Special Aids package that allows physically handicapped people to use the IIGS more easily. (Actually, CloseView isn't an NDA, though it does appear in the menu.)

*CPU Use**

The CPU Use NDA is a simple tool for showing you how much of your computer's capacity is being used by the program you're using. It's a well known fact that computers spend much of their time waiting for you, the user, to do something. And the faster the computer, the more time it spends waiting.

Just choose CPU Use from the menu for a bar graph display (Figure 8C) that's updated every half a second. You can even leave CPU Use open while you work to see just how much CPU activity the things you do generate.

CPU Use is freeware by Bill Tudor.

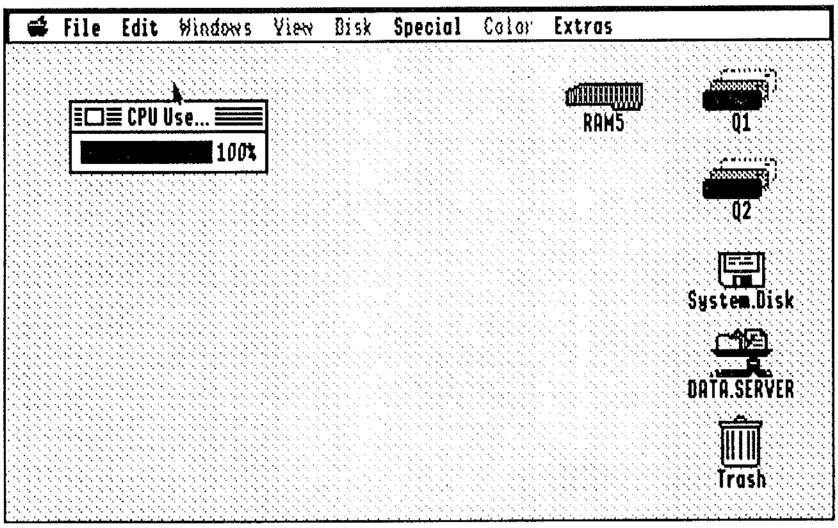


Figure 8C—CPU Use NDA

*Enigma**

Enigma (Figure 8D) is a game of logic for one player in the tradition of Mastermind. The object of the game is to use the process of elimination to arrange colors in the grid to match a secret code sequence that has been randomly chosen by the computer.

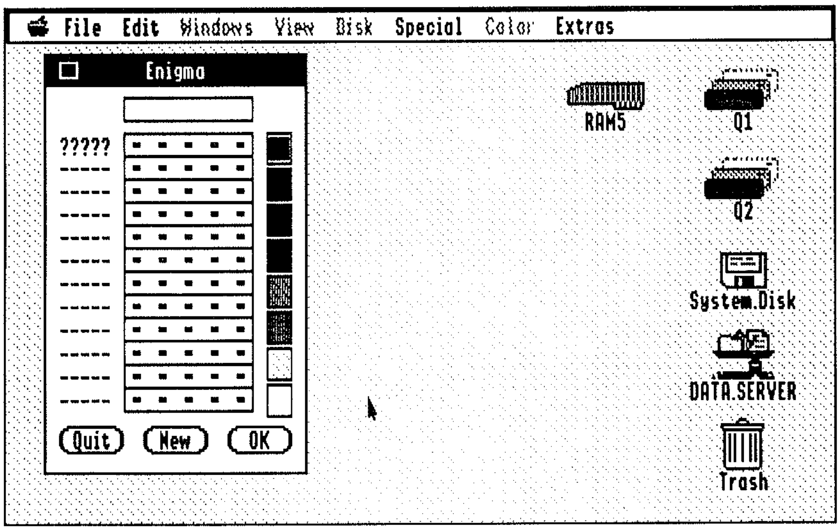


Figure 8D—The Game of Enigma in an NDA

When you open Enigma from the menu, you'll be ready to start a game. The empty rectangle at the top of the window is where the secret code will be revealed. There are twelve rows of five pegs each where you'll put your guesses. To the left of each row are five characters that tell you the results of your guess. The guess you're currently working on has "?????" next to it.

Click a color followed by a peg to place a color on that peg. When you've placed an entire row of colors, click "OK" to reveal the computer's clue. If you've picked any of the correct colors in the secret code, an "0" will appear in the clue for each correct color chosen. If you've picked a correct color and placed it in the correct position, an "X" will appear in the clue for each color you've placed correctly. (When you get five "X"s, you've solved the puzzle and the correct answer is revealed).

If you use up all twelve rows and still don't solve the puzzle, you've lost and Enigma displays the correct answer. To start a new game, click "New;" to quit the current game and reveal the answer, click "Quit."

Enigma is from the GS Desk Accessories package, © 1989 Beagle Bros, Inc.

*File Manager**

File Manager (Figure 8E) is a sort of "mini Finder" that is available from within all your programs. It allows you to move, copy, delete, and view the contents of files; create new folders; view and modify files' Icon Info; format, erase, and rename disks; and find files based on a partial name (much like the Find File NDA).

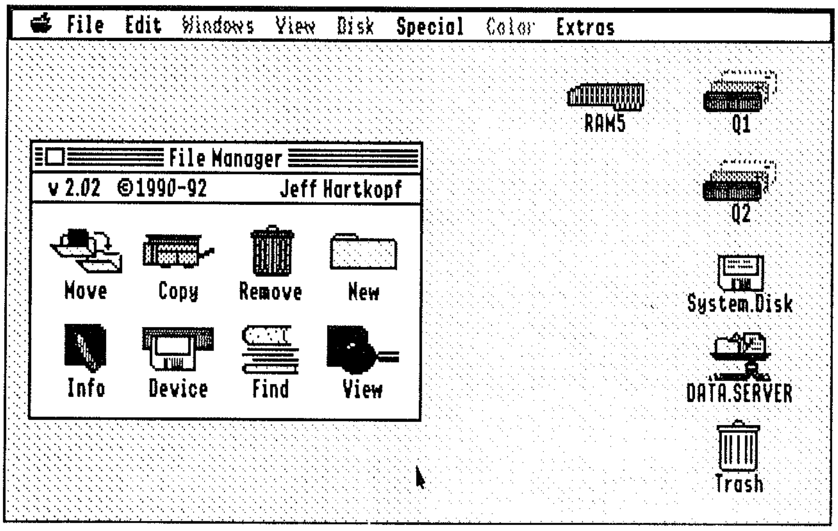


Figure 8E—The File Manager NDA

- *Move:* Click the Move icon and a Standard Open dialog will appear to select the files or folders you wish to move. (You can select more than one.) Click Accept and a Standard Save dialog will appear to let you select the target directory for the file being moved and the file's new name. Click Save to begin the Move operation. If you move the file to the directory it's already in but use a different name, the file will be renamed. If you move the file to a different disk, it will be removed from the source disk. Each file you have chosen to move presents a different Save dialog, so you can easily move files to entirely different folders.

- *Copy:* Click the Copy icon and a Standard Open dialog appears. Select the files or folders you wish to copy. As with Move, you can select more than one. Click Accept and a Save dialog appears. Select the target directory and the file's new name, then click Save to begin copying.

- *Remove:* Allows you to remove a file from a disk, just like dragging it to the Trash. The difference is that the file is removed immediately—it doesn't go to the Trash first. Choose the files to be removed from the Standard Open dialog

- *New:* Allows you to create a new folder. Use the Standard Save dialog to move to the folder you want to create the new folder in, then enter its name and click the "New Folder" or "Save" button. (Since System 6 has a New Folder button built into all its Standard Save dialogs, you may not use this feature very much anymore.)

- *Info:* Allows you to view and change a file's Icon Info. Use the Display pop-up menu to view the file's size, type, access permission, and creation/modification date. Most of these displays also allow you to change the file's attributes—something the Finder doesn't allow—but you should be very careful in doing that! Changing a file's type usually makes it useless for its intended purpose, for example. Use the "Next" button to view the next file you've selected.

- *Device:* Allows you to initialize, erase, and rename disks. As you click a device name in the dialog, the corresponding disk name appears in the "Name" line-edit field (the field is blank if the disk is unformatted or if no disk is in the drive). Type a new name and click "Initialize," "Erase," or "Rename."

- *Find:* Allows you to find files based on a partial or complete name. Unlike the Find File DA, you can select to search a folder instead of searching an entire disk via the Standard Open dialog, and you also can tell File Manager whether to ignore upper-lower case differences . However, File Manager won't search in the background, and it displays found files by their pathnames.

- *View:* Allows you to view the contents of files. This is most useful for text files, but you can view any file (most files will look like garbage). After selecting a file using the Standard Open dialog, you can scroll through the file, search for a specific string, and select and copy contents of the file to the clipboard for pasting to another document.

We've only touched on the features of File Manager. It's a very feature-rich program that's sure to make using System 6 easier. For further information, consult the documentation file "FileMgr.Docs" in your Desk.Accs folder.

File Manager is shareware © 1992 Jeff Hartkopf. The usual shareware fee is $20. Through special arrangement with the author, owners of the System 6 Bonus Pack can use the included version of File Manager without paying the shareware fee. This applies only to this version of File Manager; for future upgrades, payment of the new version's shareware fee will be required. If you register the current version, you'll receive a complete printed manual and upgrade notices for future versions.

Find File

Even the best organization sometimes falls flat on its face. You *know* you saved that file somewhere, but it's not where you expect it to be. Not to worry—the Find File NDA (Figure 8F) can find it, as long as you remember part of the name.

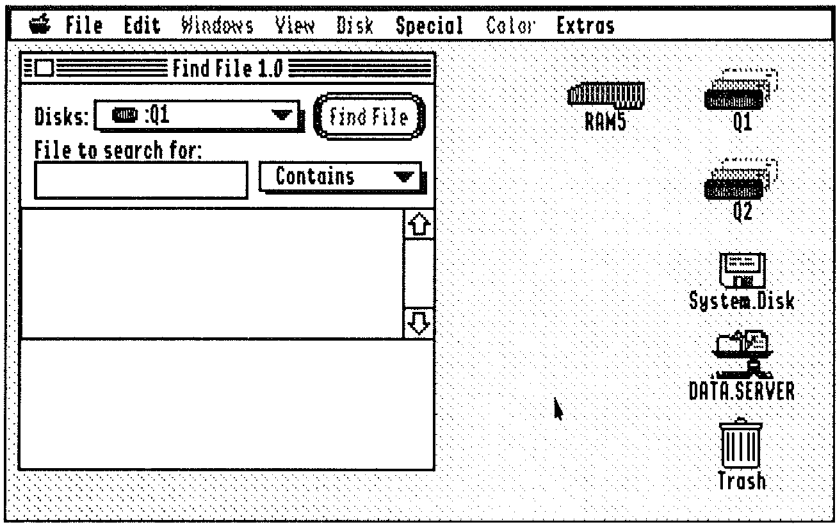


Figure 8F—Find File NDA

Use the pop-up Disks menu to tell Find File which disk to search. Type however much of a file's name as you can remember—usually a few letters or one word will be enough—into the line-edit field. Use the pop-up menu to the right of the line-edit field to tell Find File whether you think the name of the file *contains* what you typed (the most general search), *matches* it exactly (the most specific search), or *begins* or *ends* with what you typed. Then click the "Find File" button and the search begins.

You can continue working while Find File works. Just bring your document window to the front and Find File will continue searching in the background. (It searches faster while the Find File window is in front.)

When Find File finds a file that matches the search criteria, it displays its name in the list just below the line-edit field. If you've chosen a phrase that occurs a lot on your disk (try searching for all files that contain the letter E, for example!), you'll end up with a long, scrollable list. You can scroll this list while Find File continues searching—it takes the system a second to acknowledge a mouse click while Find File is working.

Click any file on the list to find out where it is (Figure 8G). In this example, we've clicked the file SCSI.Manager. The folder list (below the file list) tells us the file is in the Drivers folder, which is in the System folder, which is on the volume Q1.

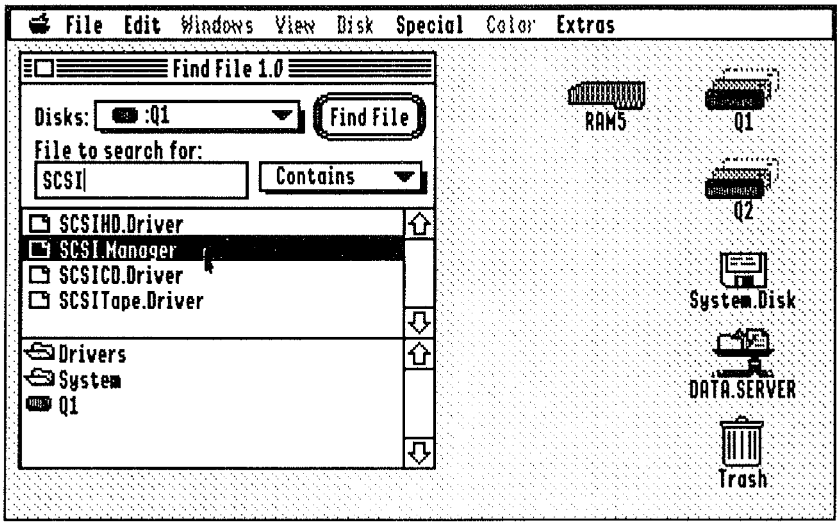


Figure 8G—Checking the Found Files List

When Find File finishes searching, you can continue looking at the locations of the files that Find File has found, or prepare another search by typing a new search word.

*Key Find**

Most IIgs fonts support extra characters that can be typed using the Option key. Most of these are foreign characters and symbols. You may occasionally find these useful to include in your document, but it's not always easy to remember that Option-2 produces the "™" character. Key Find (Figure 8H) to the rescue!

Click on any character in Key Find's grid and Key Find displays the keystrokes necessary to produce that character (for example ñ is Option-N followed by a lowercase "n"), as well as a large-size view of that character. Use the Copy command (⌘-C) to copy the character to the clipboard, then return to your document and paste it in.

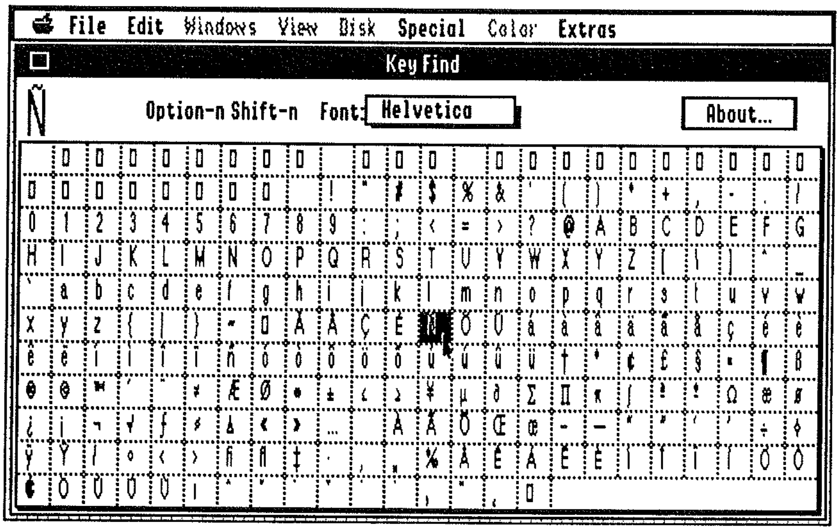


Figure 8H—KeyFind NDA

The Font pop-up menu lets you determine which font is displayed inside the grid—sometimes different fonts have different special characters. There's only one quirk—it takes a second or two for Key Find to be ready after you first click on the grid. Keep pressing and eventually Key Find will spring to life.

Key Find is freeware by Alfredo Velella.

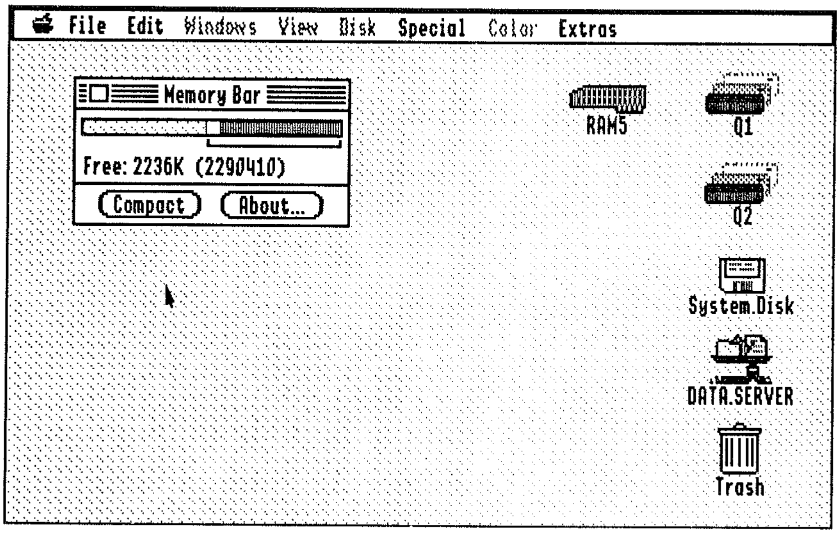


Figure 8J—Memory Bar NDA

*Memory Bar**

The Memory Bar NDA (Figure 8J) makes a good companion to the CPU Use NDA. In graphic form, it shows you just how much of your computer's memory is in use and how much is available for use. Unlike the Finder's "About The Finder," the Memory Bar NDA is available within all Desktop applications, because it's an NDA.

The yellow portion of the bar represents the amount of memory in use. The green portion represents the amount of memory that is in use but will be released if another program needs it (purgable memory). The pink portion indicates the largest *contiguous* block of memory available—how much memory a program can allocate all at once (MaxBlock). And the white area represents how much memory is not in use but is not included in MaxBlock (the white and pink bar together represent free memory).

To find out exact memory sizes for any of the bars, click on the bar you're interested in or press the numbers 1-5. You can also use Tab to cycle through the available bars.

Click the "Compact" button to get rid of as much purgable (green bar) memory as possible and make it available for reuse by the system. Usually you won't need to use the "Compact" button because the system will give up purgable memory as it's needed. However, you may find this button useful before launching some memory-hungry programs.

☛ Warning!

Don't compact memory while you're running AppleWorks GS. Evidently that program marks part of its memory as purgable when it really isn't Compacting memory is a good way to crash AppleWorks GS.

Memory Bar is public-domain software by Dave Lyons.

*Minstrel**

Minstrel (Figure 8K) allows you to play SoundSmith and SynthLab songs in Desktop programs while you continue to work! It's like a jukebox for your IIGS. SynthLab is a program included with System 6 for recording and playing back high-quality music. Now you can listen to those songs any time you like.

Minstrel keeps track of a list of songs in its "jukebox." To add a song to its jukebox, click the Insert button (you'll have to do this first because there are no songs in the jukebox to begin with). Using the Standard File dialog, open a SoundSmith or SynthLab song. It appears in the list. Add any other songs you might want to listen to, too. You can add as many as you want from one folder—hold down the Shift key while clicking to select all the files from the current selection to the click location; use the ⌘ key while clicking to select and unselect individual files. Click Accept to add the files you selected. (To remove a song from the list, click "Eject.")

To play a song, select it from the list and click the "Play" button. To play all the songs in the list in a random order, click "Shuffle." To repeat the currently playing song, click the "Repeat" button. The "Rewind," "Forward," and "Pause" buttons are functional during playback. "Stop" works just as you'd expect.

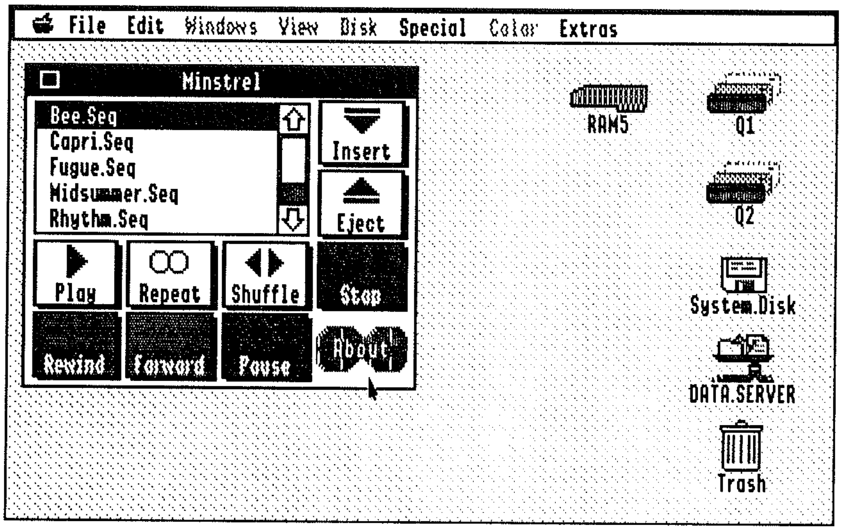


Figure 8K—Minstrel, the Music NDA

Minstrel will continue playing songs even if you bring your document to the front and resume working. Your main program will run somewhat slower while Minstrel is playing sounds. (An accelerator helps.)

Minstrel, by Chris McKinsey, is copyright © 1991 by Softdisk G-S, a monthly disk publication. Used by permission. For subscription information, call 1-800-831-2694 Ext. 1005.

*Note Pad**

Ever wish you had a place for jotting notes to yourself? If you're like us, your desk is covered with little tiny slips of paper with things scribbled on them—or worse, those sticky Post-It notes. The Note Pad (Figure 8L) is an electronic solution to the clutter problem. Just type your notes into the Note Pad and they'll always be handy in any Desktop program.

The Note Pad contains eight numbered pages for your jottings. To move from one page to the next, click the upturned page corner in the lower left . To move to the previous page, click the edge of the page showing under the upturned corner.

Note Pad is from the GS Desk Accessories package, © 1989 Beagle Bros, Inc.

*ScientCalc**

If you prefer a full-featured scientific calculator to Apple's functional but wimpy four-banger, try ScientCalc (Figure 8M). It has all the functions of a TI-30 or similar calculator, including memory, factorial, hexadecimal and binary functions, logarithms, trigonometric functions, and even something no real calculator we've ever seen has—built-in help!

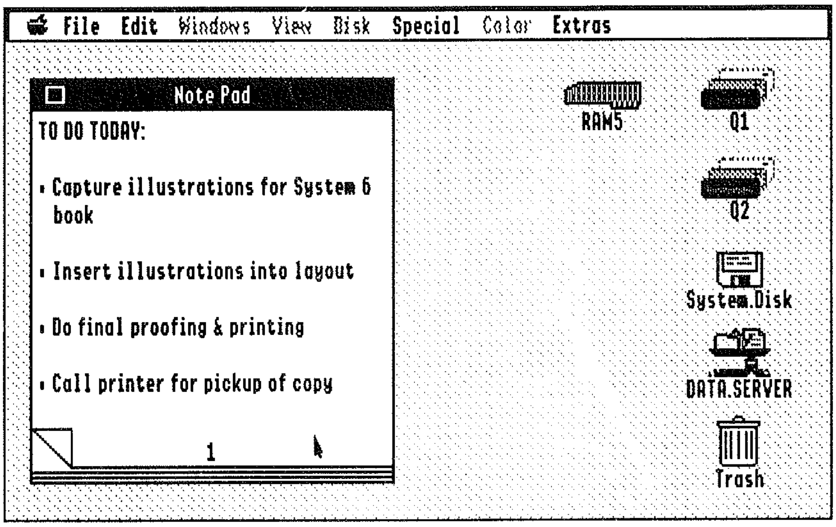


Figure 8L—The Note Pad NDA

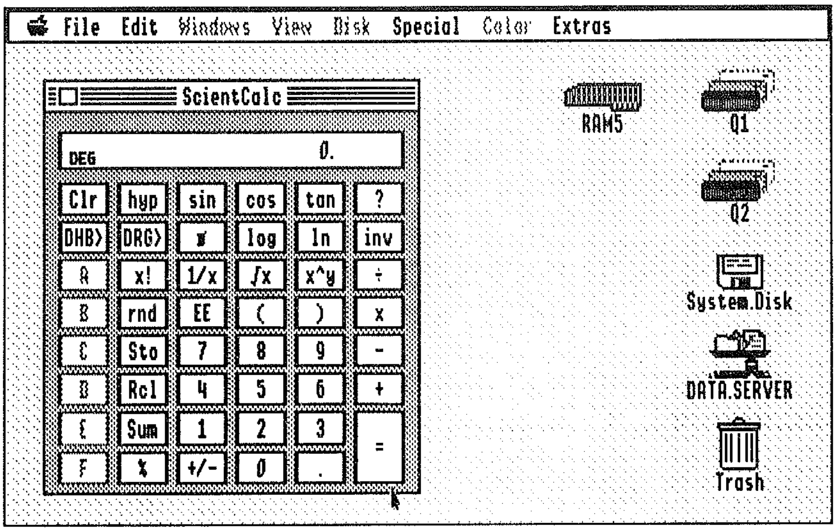


Figure 8M—The ScientCalc NDA

If you're familiar with scientific calculators, using this one should be a breeze. The built-in help (click the "?" button) lists which buttons do what, and which keys on the keyboard activate buttons on the keyboard (most buttons can be activated from the keyboard). We've also included a documentation file that you can read with Teach—look in the Desk.Accs folder.

ScientCalc (also known simply as Calculator) is shareware © 1992 Jeff Hartkopf. The usual shareware fee is $15. However, through special arrangement with the author, owners of the System 6 Bonus Pack can use the included version of ScientCalc without paying the shareware fee. This applies only to this version of ScientCalc; for future upgrades, payment of the new version's shareware fee will be required. If you register the current version, you'll receive a complete printed manual and upgrade notices for future versions. ***Note:*** *We changed the name to ScientCalc to avoid conflict with Apple's Calculator, which has the same name. However, the rest of the NDA is the same as the shareware version.*

*Scrapbook**

The clipboard is useful for copying information from one application to another, but it only holds one item (text or graphics) at a time. The Scrapbook (Figure 8N) gives you a place to store all your clip art and text where it can be accessed easily. Use the scroll bar at the bottom of the Scrapbook window to view your clips.

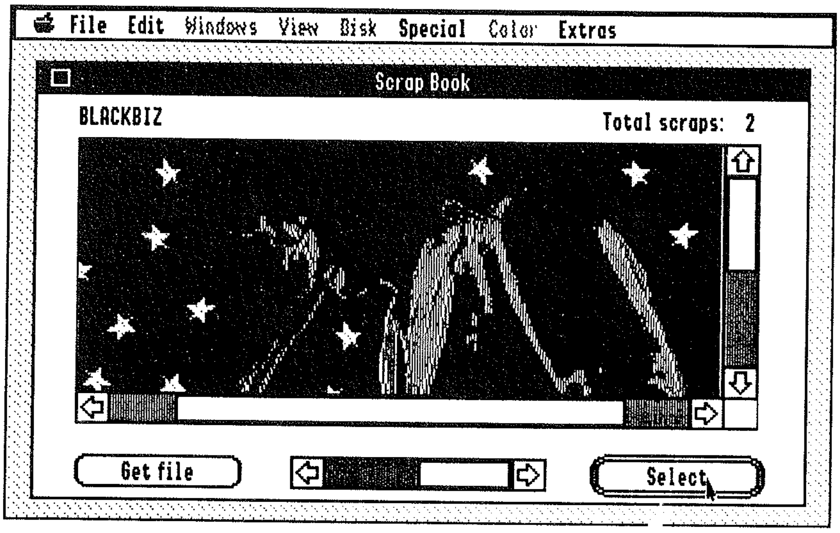


Figure 8N—The Scrapbook NDA

To store text or graphics into the Scrapbook, just use the "Cut" or "Copy" commands to place the desired material on the clipboard from the desired application. Then open the Scrapbook (or bring it to the front if it's already open) and select "Paste" (or press ♂-V).

You can also use the "Get File" button to add a clip to the Scrapbook. You can place any standard (Apple Preferred format) graphics file, or any standard text file, on the clipboard this way. A Standard Open dialog will appear; select the file you want to use and the entire file (up to 32K of graphics or 16K of text) will be placed on the Clipboard. Then paste it into the Scrapbook using ⌘-V.

To copy text or graphics from the Scrapbook into an application, use the scroll bar to view the clips until you find the one you want to use. (Or use the Select button to bring up a Standard Open dialog to allow you to select the scrap by name.) Then select "Copy" (or press ⌘-C), switch back to your application, position the insertion point where you want the scrap to be inserted, and select "Paste" (or press ⌘-V). You can also select part of an image in the scrapbook by dragging a rectangle around the section of the image you want to use before Copying it.

Scrapbook is from the GS Desk Accessories package, © 1989 Beagle Bros, Inc.

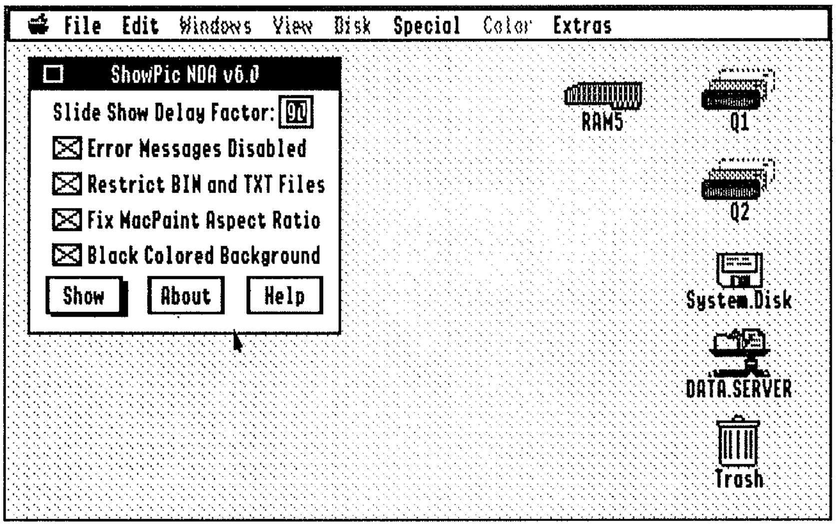


Figure 8P—ShowPic Graphics Display/Conversion NDA

*ShowPic6**

ShowPic6 (Figure 8P) lets you view virtually any graphics you may encounter, no matter what format they may be in. ShowPic can load and display:

- Unpacked graphics (Screen) files
- Eagle/Packbytes format graphics
- Apple Preferred Format graphics
- PaintWorks (320/640) format graphics
- Print Shop GS color graphics
- Normal & packed 3200-color pictures (APP & APF)
- Graphics Interchange Format (GIF) 87a & 89a
- MacPaint graphics

Just click the Show button. A Standard Open dialog will appear; select the files you want to display and click "Accept." To save a displayed graphic in screen format (which can be read by most paint programs), press ⌘-S and use the Standard Save dialog to specify the name and the target directory for the file. ShowPic functions both as a simple "slide show" program (adjust the time delay in the line-edit field) and a simple graphics converter.

There's a lot more, too. Unfortunately, we don't have the space to go into it. Click the "Help" button for more information; we've included a documentation file (called "ShowPic.Docs") as well. (This documentation file is placed in your Desk.Accs folder when you install the Bonus Pack Desk Accessories.. Double-click it to open it in Teach.)

ShowPic is freeware by Dave Leffler and Lunatic Jonathan Bruce E'Sex.

Finder Extensions

As we've mentioned, Finder extensions are desk accessories for the Finder. There's one included with System 6 and two more in the System 6 Bonus Pack.

EasyMount

EasyMount is used to create small documents that automatically mount network volumes when double-clicked in the Finder. Just click the shared disk you wish to create an auto-logon document for, then choose "Create server alias" from the Extras menu. The EasyMount server alias can then be double-clicked to log onto that network volume.

IR*

IR is a Finder extension developed by Matt Deatherage at Apple. Originally, IR stood for "Init Restarter" and was used to load init files (files usually loaded by the system during startup) *after* the system had already been started. It has evolved into a tool for loading NDAs, CDAs, drivers, inits, and Finder extensions—all of which *usually* must be loaded at boot time—just by double-clicking them in the Finder.

Okay, that's what IR does—but what does it *do?* In practical terms, having IR in your system means that you don't have to keep infrequently used system elements (particularly DAs) in your System folder all the time. (Having lots of desk accessories in your system folder means less memory for your programs and longer boot times.) Even though these DAs don't normally appear in your menu, you can double-click them in the Finder and they'll open like magic—and automatically be added to your menu until you restart your computer. The same is true of drivers (you might leave your 5.25" driver out of the System folder if you rarely use 5.25" disks, then double-click it if you decide you want to use it in a Finder session—voila, your 5.25" drives appear on the desktop) and Finder extensions. Even inits—like Apple's GSBug debugging utility (not included) and the System 6 Special Aids—can be installed with IR.

☞ **Note!**

There are few DAs and Finder extensions that don't like to be opened up by IR. All the ones in the Bonus Pack work fine, but be sure to try IR with other new DAs before relying on it to work. Some of them just won't—that's not a bug in IR, it's just a fact. We decided that even so, IR is more than useful enough to include in the Bonus Pack.

IR also has a Preferences item on the Extras menu (Figure 8Q). Check "Install NDAs instantly" if you want IR to install NDAs in the menu immediately (if you uncheck this box, the menu isn't updated until you launch an application). "Open NDAs if possible" tells IR whether or not to automatically open NDAs after they're installed (if not, they're only installed in the menu).

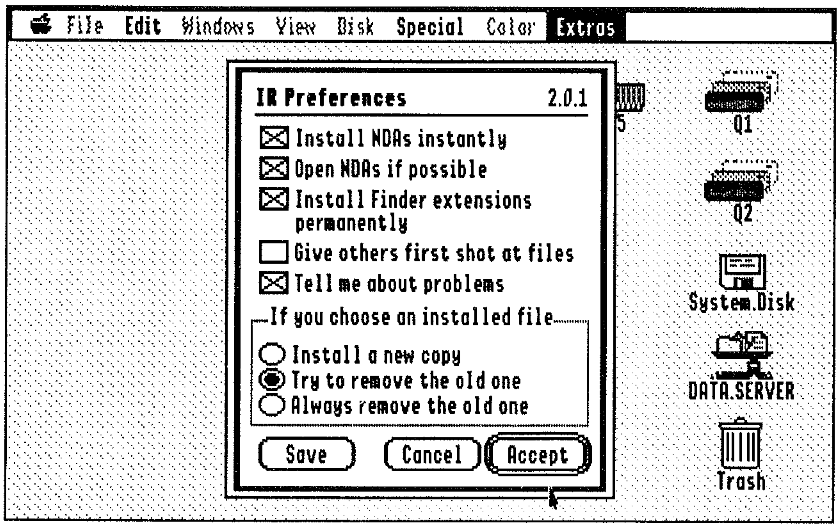


Figure 8Q—IR (Say "Ear") Preferences

"Install Finder extensions permanently," if checked, tells IR to install Finder extensions for the remainder of the session (until you restart); if unchecked, Finder extensions go away when you leave the Finder. "Give others first shot at files" allows applications to see these system files (as documents) when they're double-clicked; only if there's no application for those files will IR open them. "Tell me about problems" tells IR to put up alerts when errors occur.

The radio buttons let you tell IR what to do if one of the things you're installing is already there—install a new copy, try to remove the old one (if that's not possible, install the new one anyway), or always remove the old one (if that's not possible, do nothing).

When you're done setting preferences, click "Accept" (to save the changes in memory until you restart) or "Save" (to save the changes permanently to disk). "Cancel" dismisses the dialog without saving your changes.

Shortcut

The following keys activate various buttons in the IR Preferences dialog:

- ⌘-N Toggle "Install NDAs Instantly"
- ⌘-F Toggle "Install Finder extensions permanently"
- ⌘-I Select "Install a new copy"
- ⌘-T Select "Try to remove the old one"
- ⌘-A Select "Always remove the old one"
- ⌘-S Click "Save"

We've included the original documentation written by Matt Deatherage. Look in your System.Setup folder for "IR.Docs" after installing the Bonus Pack.

IR 2.01 by Matt Deatherage © 1992 Apple Computer, Inc. Licensed for use with the System 6 Bonus Pack.

*Quick Launch**

Quick Launch (Figure 8R) is an Finder Extension that lets you add your most frequently used applications to the Extras menu. No more digging through your folders or moving windows out of the way of the desktop items to launch applications! Select "Quick Launch list" from the Extras menu to tell QuickLaunch which applications to put in the Extras menu.

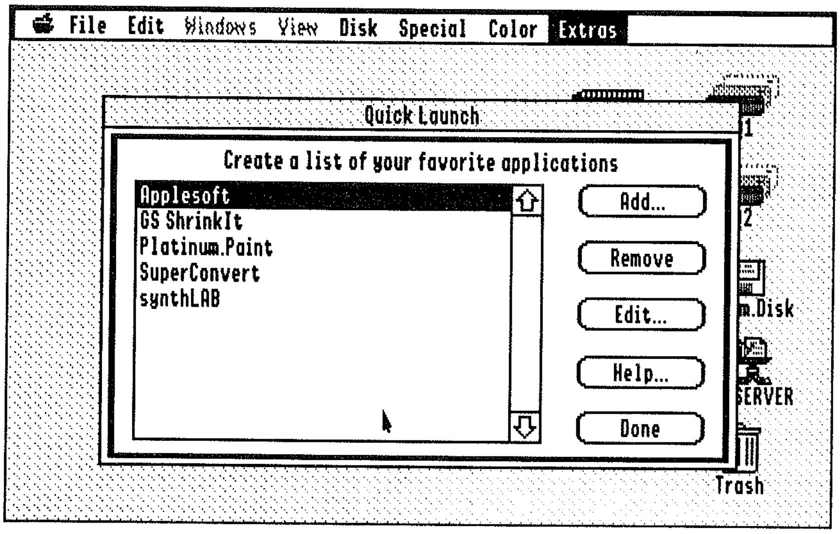


Figure 8R—QuickLaunch Configuration Dialog

Click the "Add" button and a Standard Open dialog will appear. Select the application you want to add to the menu and click "Open." Then click "Done" and check the Extras menu. Your new application should be listed just under "Quick Launch list" with a tiny application icon to its left. Removing an application from the list is just as simple—click the item in the Quick Launch dialog, then click "Remove."

You can also edit your applications once they've been added. For example, if you add Aplworks.System to the list, you can edit the item on the menu to read "AppleWorks"—in plain English. You can also assign a keystroke to launch that application: in the Menu keys line-edit field, just type the character you want to use. (If you're using a letter, "A" for example, type "Aa" to assign both upper and lower-case versions of the letter to the application.) Then you can press, for example, ⌘-A to launch the application instead of pulling down the menu.

Full online help is available. Click the "Help" button and choose an appropriate topic from the "Topic" pop-up menu. When you're done setting up your application list, click "Done."

Chapter Nine

WORKING WITH CONTROL PANELS 9

About Control Panels

You're probably familiar with the built-in IIGS Control Panel. It's a Classic Desk Accessory and is accessed by holding down ⌘ and Control while pressing and releasing Escape, then selecting "Control Panel" from the CDA menu. (If you're not familiar with the built-in Control Panel, grab your IIGS Owner's Manual and take a few minutes to learn about it.)

You use the built-in Control Panel to view and change the various settings of your IIGS—how fast the keys repeat on the keyboard, whether a particular slot in the computer is connected to a card or to a port on the back panel, and things like that. The built-in Control Panel is always available from the instant you turn on your computer, and it remains available in most programs (they don't have to be Desktop programs).

System 6's control panels perform the same function, except that they only work in Desktop programs. Instead of having *one* big desk accessory for all functions (like the built-in Control Panel), there's a control panel for each function—one for choosing a printer, one for adjusting the functionality of your keyboard, one for adjusting your slot mapping, and so on.

⑥ New & Improved

In Systems previous to System 6, the NDA itself was called the Control Panel, and each control panel in it was called a Control Panel Device, or CDev for short. Each CDev appeared in the same window (part of the Control Panel NDA window) so only one CDev could be open at a time. In System 6, each control panel appears in its own window; the new Control Panels NDA is just a convenient way of opening the control panels (they can now be opened by the Finder).

▲ By The Way...

The terminology change from "CDevs" to "control panels" parallels a similar change recently made on the Macintosh, when Apple released Macintosh System 7.

Control panels are stored in a folder called CDevs inside the System Folder. Installing new CDevs is as simple as dragging them into this folder. Usually, control panels can be stored in other folders, too, but there are a few control panels that definitely need to be in the CDevs folder because they do something at startup time.

Opening Control Panels

There are two basic ways to open control panels in System 6. First, as depicted in Figure 9A, you can open them with the Finder by opening the CDevs folder (inside the System folder) and then double-clicking the desired control panel.

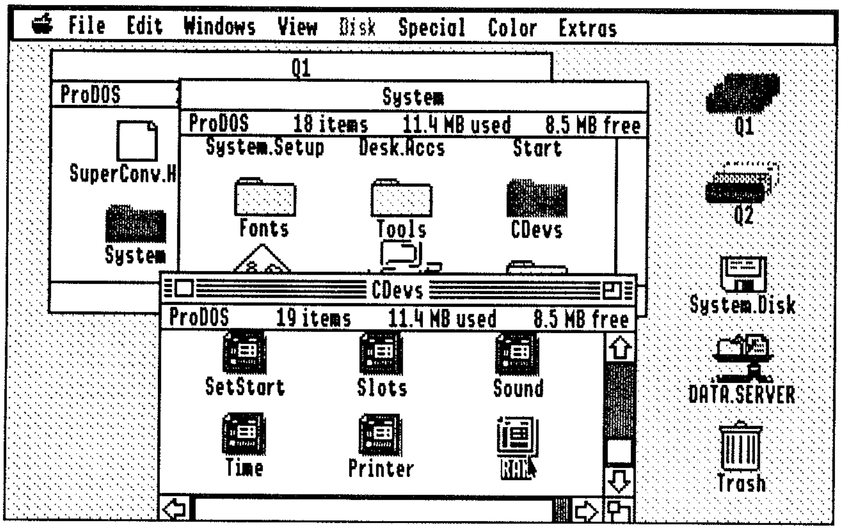


Figure 9A—Opening a Control Panel with the Finder (One Way To Do It)

Shortcut

Holding down ⌘, Option, and Shift while pressing and releasing Escape opens the CDevs folder on the startup disk, without having to open the other two windows to get to the CDevs folder. Another popular shortcut is to drag one or more control panels out of the CDevs folder and onto the desktop. (They'll still appear in the Control Panels NDA, too!)

Double-clicking a control panel, obviously, only works when you're in the Finder, and it'd be rather inconvenient to have to quit the program you're using every time you wanted to change some aspect of your IIGS's functionality. Therefore, System 6 also includes a Control Panels NDA (Figure 9B) that can be accessed in any Desktop program.

Shortcut

Hold down ⌘ and Shift while pressing and releasing Escape to pull up the Control Panels NDA. (This is analogous to the ⌘-Control-Escape sequence used to access the Control Panel CDA.)

Scroll through the list of installed control panels, then double-click one to open it. (Alternately, you can use the "Open" button to open the highlighted control panel, or click the "Help" button for more information a control panel).

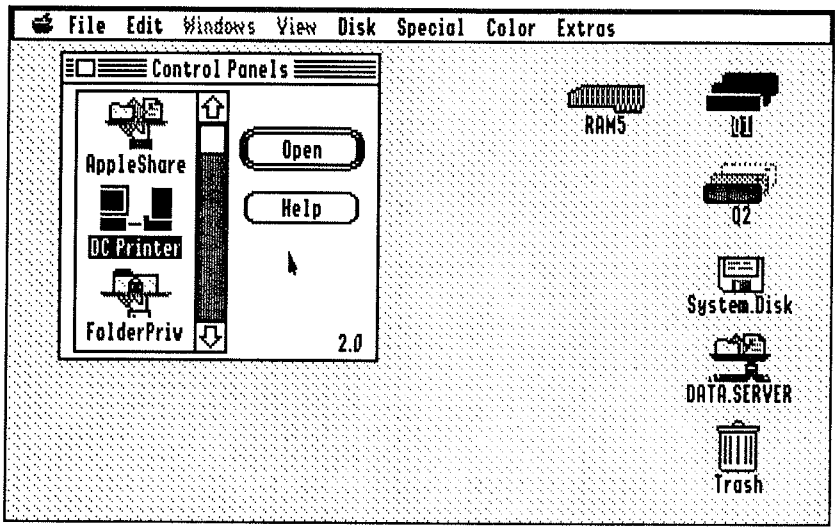


Figure 9B—The Control Panels NDA (The Other Way To Do It)

⑥ New & Improved

In System 5's Control Panel NDA, you only needed to click the icon of the desired CDev once to activate it. Apple decided that this was inconsistent—CDevs are displayed as icons, and you should double-click icons to open them. Thus, in System 6 you double-click.

Shortcut

You can also use the arrow keys to select a control panel in the Control Panels NDA, and press Return to open the selected control panel. Typing the first few letters of a control panel's name also works to select it.

With the Control Panels NDA, you can open as many control panels as you like. Control panels even stay open when you close the Control Panels NDA. Control panel windows can be moved around wherever you like on the screen, and you can even leave them open while you work by clicking a document window and bringing it to the front.

The rest of this chapter is a reference describing each control panel included with System 6.

DC Printer

The DC (for Direct Connect) Printer control panel (Figure 9C) allows you to tell the system which printer is connected to your computer and how it's connected. The top list allows you to choose the printer port, the modem port, or a parallel card in slot 1; the bottom list allows you to choose an Epson, ImageWriter, ImageWriter LQ, LaserWriter, or StyleWriter printer. Click the appropriate settings for your system, then close the control panel.

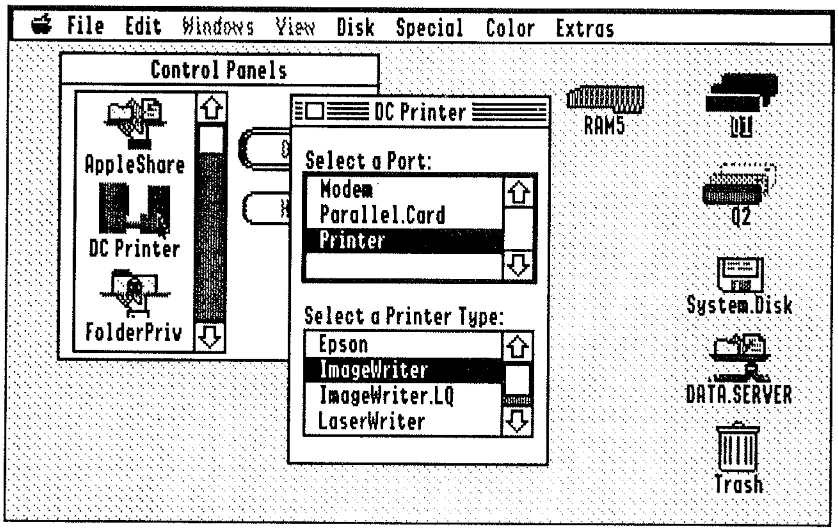


Figure 9C—The Direct Connect Printer Control Panel

Note!

The ports and printers listed will depend on which printer drivers you've installed. If you don't see your printer, check the Installer to see if it has a driver for your printer. If it does, install it. If it doesn't, investigate third-party driver packages, such as Vitesse's Harmonie and Seven Hills' Independence, which provide drivers for a wide range of printers that Apple doesn't support directly.

Shortcut

The DC Printer control panel, like many of the other control panels, can also be controlled by the keyboard. Use the Tab key to toggle between the port and printer type lists; use the up and down arrow keys (or letters on the keyboard) to move the highlight bar to a particular item.

General

The General control panel (Figure 9D) allows you to change a number of IIGS system settings, all of which are controlled by a pop-up menu. (The default, or normal, settings are displayed in *italic* type in the menu.)

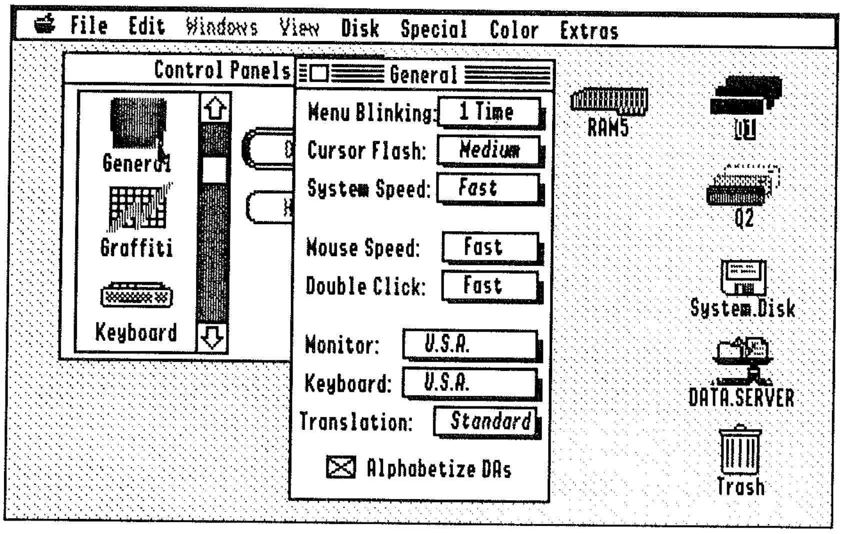


Figure 9D—General Control Panel

- *Menu Blinking:* Controls how many times an item in a pull-down menu flashes after you select it.
- *Cursor Flash:* Controls how quickly the insertion point flashes. Also controls the flashing cursor in some (but not all) older, 8-bit programs, especially BASIC ones.
- *System Speed:* We suggest keeping this set to "Fast" at all times.

- *Mouse Speed:* Controls how far you have to move the mouse to move the pointer across the screen. The faster the Mouse Speed, the *less* you have to move it. (ROM 01 IIGS owners can choose two settings here; ROM 03 owners have more choices.)
- *Double Click:* Controls how fast you must click the mouse for the computer to recognize two clicks as a double-click instead of two separate click. Set this too fast, and the computer may not recognize your double-clicks; set it too slow, and the computer may recognize a double-click when you really intended two single clicks.

- *Monitor:* Determines the display language for the screen. Most of you are 'murricans, so the default setting of "U.S.A." should be fine.
- *Keyboard:* Determines the keyboard layout. "U.S.A." should be fine for most of you.
- *Translation:* Controls keyboard translation (for foreign and special characters). If a program isn't responding properly to Option keystrokes, set this option to "None."

There's also a check box that determines whether NDAs are displayed in alphabetical order (the default) or in the order they appear in the Desk.Accs folder.

Keyboard

The Keyboard control panel (Figure 9E) lets you set attributes of your keyboard with a variety of pop-up menus and checkboxes. The options are:

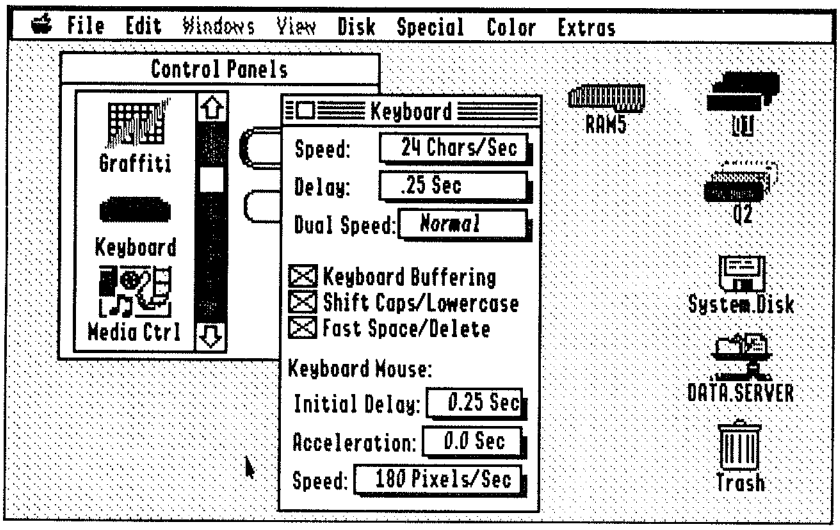


Figure 9E—Keyboard Control Panel

- *Speed:* Sets your keyboard's repeat speed, in number of characters per second. Longtime IIGS users will keep this set at its highest setting.
- *Delay:* Sets the amount of time you need to hold a key down before it begins to repeat. Again, old hands will set this to its minimum.
- *Dual Speed:* Controls whether the arrow keys, space bar, and Delete keys can be further accelerated during repeat by pressing the Control key.

- *Keyboard Buffering:* Controls whether the IIGS "buffers" keystrokes. If the computer is not responding to your keystrokes as quickly as you like, the buffering feature allows you to "type ahead" of the computer without losing keystrokes. Some older programs are incompatible with this feature. To clear buffered keystrokes before the program "sees" them, hold down ⌘ and Shift while pressing Delete.
- *Shift Caps/Lowercase:* Controls whether the Shift key produces lowercase letters when the Caps Lock key is down.
- *Fast Space/Delete:* Controls whether the space bar and delete keys repeat faster than the rest of the keys.

The Keyboard Mouse feature is available on ROM 03 IIGS machines, and on ROM 01 if the Easy Access feature is installed. (Easy Access is *not* installed by the "Easy Install" procedure.) The pop-up menus here control how long it takes the Keyboard Mouse to begin accelerating, how fast it accelerates, and its maximum speed.

You activate (and deactivate) the Keyboard Mouse by holding down ⌘ and Shift while pressing the Clear key. The numbers around the "5" key on the numeric keypad move the mouse in the obvious direction—for example, "7" moves the mouse up and to the left, and "2" moves the mouse down. The "5" key clicks the mouse button; the "0" key holds the mouse button down (for dragging) until you release it by clicking the "5" key once.

MIDI

The MIDI control panel allows you to specify which type of MIDI (musical instrument) interface you have attached to your IIGS. This Control Panel is installed when you install the SynthLab application or the Media Control package.

Modem Port & Printer Port

These two control panels (Figure 9F) are virtually identical, so we'll cover them together. We won't cover the settings in detail—they're covered in your IIGS Owner's Manual in the section on the Control Panel CDA. The most important thing to remember is that the standard settings are almost always fine for both the modem and printer ports.

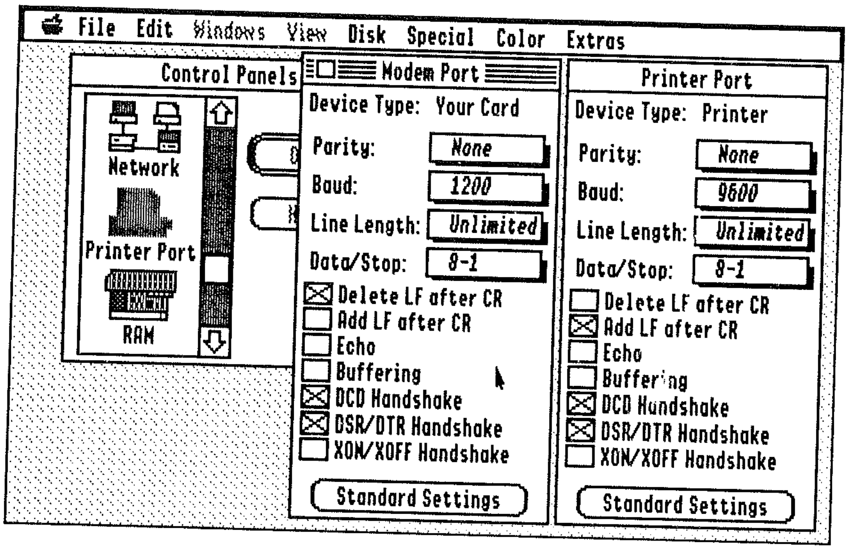


Figure 9F—Modem Port & Printer Port Control Panels

On the Printer Port control panel, you may (very rarely) need to set Baud to something other than "9600" if you've hooked up a printer other than a standard Apple printer. The "Add LF after CR" checkbox is also one to keep an eye on—if your printer double-spaces everything, try turning it off. If your printer prints everything on one line, try turning it on.

The Modem Port control panel will hardly ever need to be changed, because most modem programs set these parameters for you automatically. You don't need to set the Baud to "2400" if you have a 2400 baud modem; your telecommunications package will handle this. (The exception is when you're using the built-in modem firmware to telecommunicate; hardly anyone ever does this, but if you do, you already know what all these settings do anyway.)

☞ **Note!**

The America Online software requires nonstandard settings for both the Printer Port and the Modem Port. It's the only software we know of that does, and the manual for the software tells you how the ports need to be set.

If you plan to attach a modem to the Printer Port, or a printer to the Modem Port, you'll first need to use the Slots control panel to tell the IIGS that the appropriate slot should be connected as the right kind of port. Then open both the Modem and Printer Port control panels (as we've done in Figure 9F) so that you can easily set the parameters of one to match the other.

Monitor

The Monitor control panel (Figure 9G) tells the IIGS what kind of monitor is attached to your computer and the various attributes of that monitor. As usual, the default (or factory-normal) settings are displayed in italic type. The pop-up menus are:

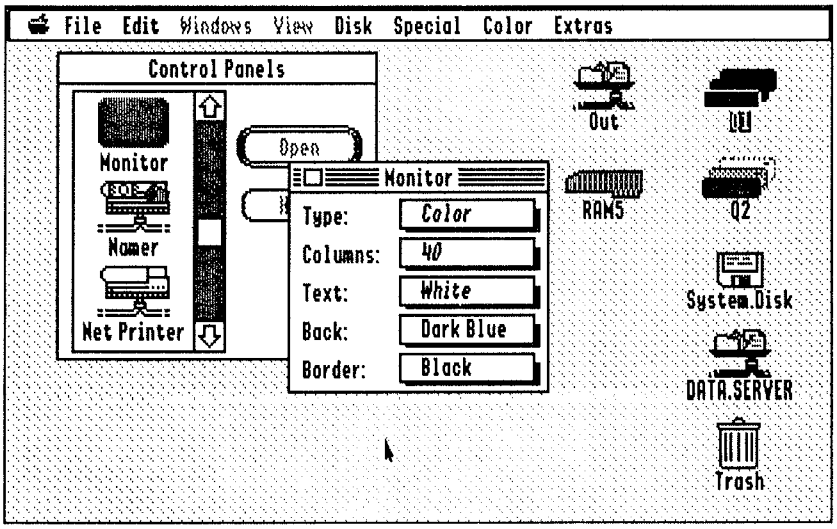


Figure 9G—Monitor Control Panel

- *Type:* If you have an RGB monitor, leave this set to "Color." If you have a monochrome (green or amber) monitor, set it to "Monochrome." You may occasionally find it useful to switch this setting if the screen is unreadable in one mode or the other. (If you have an RGB monitor, this setting only makes a difference in one place—the double-hi-res graphics mode, used by Publish It! and a few other programs.)
- *Columns:* Controls the number of columns the computer reverts to after a reset (or at power-up). We strongly recommend leaving this item set to "40." This does not mean that 80-column programs (like AppleWorks) won't work—they will.
- *Text:* Selects the color of the text.
- *Back:* Selects the color of the background. Naturally, this can't be the same color as the Text color.
- *Border:* Selects the color of the border.

RAM

The RAM Control Panel (Figure 9G) allows you to set the size of your RAM Disk and cache. In addition, it displays the same kind of information on memory usage as the Finder's "About the Finder" window and the Memory Bar NDA included with the Bonus Pack.

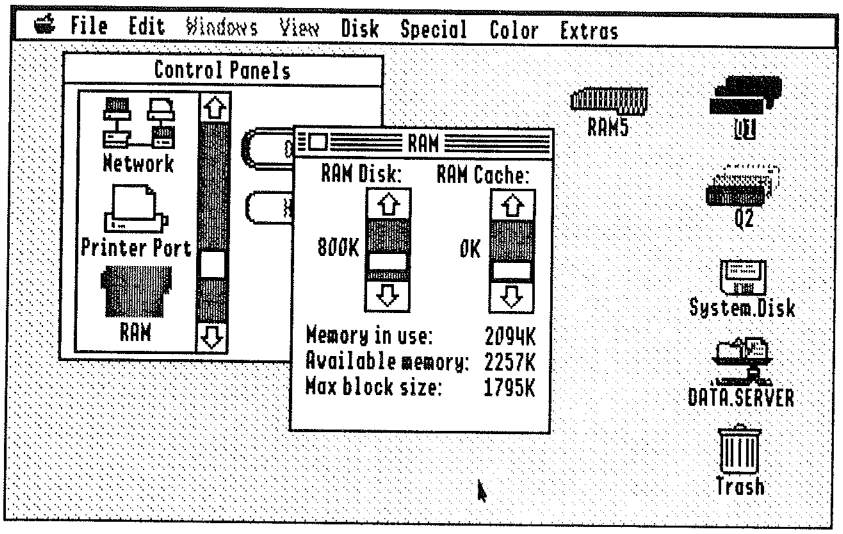


Figure 9H—RAM Control Panel

The RAM Disk is a IIGS feature that acts as a very fast, but volatile, disk drive. "Volatile" means that the contents of the RAM Disk vanish when the power is turned off. By setting the RAM Disk to 800K (if you have enough free memory to do so), you can perform a fast disk-to-disk copy of the System Disk (or any disk you use a lot) when you first start the system, then restart from the RAM Disk for accelerated access.

▲ **By The Way...**

The FlashBoot utility included with the Bonus Pack makes this even easier, and allows you to painlessly save the contents of the RAM Disk to 3.5" disk. See the FlashBoot manual.

The RAM Cache keeps the most frequently used information from your disks in memory, so that when you access your disks, a lot of the information the computer needs is already in memory. This can definitely speed your disk access when it works properly. It's usually not a good idea to allocate more than 128K worth of cache, though—larger sizes are self-defeating, since the computer spends more and more time looking to see if the block is in the cache before it looks on the disk. (You might want to try various cache sizes and time the activities you perform most often. If the cache doesn't seem to help much, release the memory.)

The cache doesn't really use the memory it uses. (Say what?) What we mean is, the memory the cache uses is purgable, so if it's needed by another program, the other program will be able to take it. The cache will not cause out-of-memory errors.

▲ **By The Way...**

Even if you set the Ram Cache to zero K, the system still maintains a small (16K) cache. According to Apple engineers, this is the smallest setting which provides acceptable performance with 3.5" drives.

SetStart

The SetStart control panel (Figure 9J) determines which program your computer starts up into. The pop-up menu offers: Finder (the default), the current application (open SetStart while you're using application you wish to start up into and choose this option), or another application.

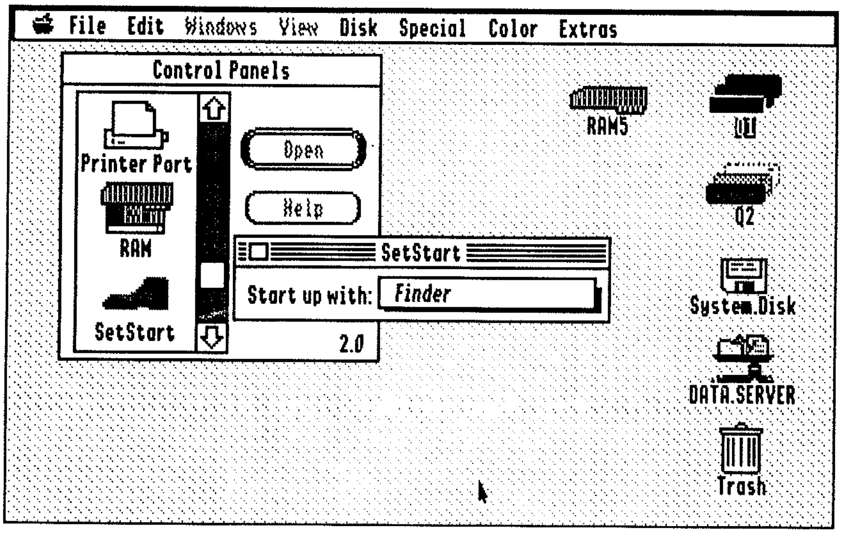


Figure 9J—SetStart Control Panel

If you choose "Select application," a Standard Open dialog appears to allow you to choose the startup application.

Note!

Even if you start up into some other application, the Finder is still considered the "previous" application when you quit the startup application.

Shortcut

If you occasionally want to start up into the Finder instead of the startup application you've selected with SetStart, hold down the ⌘ key while your computer starts up. When the SetStart icon appears with a red "X" drawn through it, release the ⌘ key. The Finder will start up instead of your normal startup application. The next time you start the computer, the application you selected in SetStart will appear.

Slots

The Slots control panel (Figure 9K) works the same as the Slots option in the Control Panel CDA. There are pop-up menus for each slot, plus another pop-up menu to decide which slot the computer starts up from. We won't cover the settings of these items in depth; they're covered in the Apple IIGS Owner's Manual. If you've used the Control Panel CDA to set your slot status and startup slot, you'll feel right at home here.

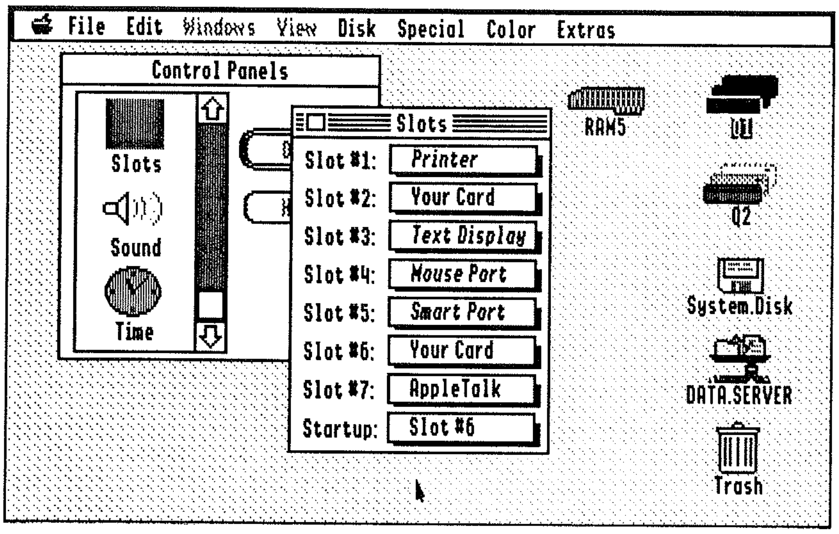


Figure 9K—Slots Control Panel

Sound

The Sound control panel (Figure 9L) allows you to set the volume and pitch of your IIGS's beep, plus it allows you to select a number of other fun sounds for your normal system beep and a number of other events. (The Sound control panel is not installed by the Easy Install. If you have a hard drive, you'll definitely want to install it.) The System 6 Bonus Pack also includes over fifty other fun sounds that you can assign to events in the Finder and other programs.

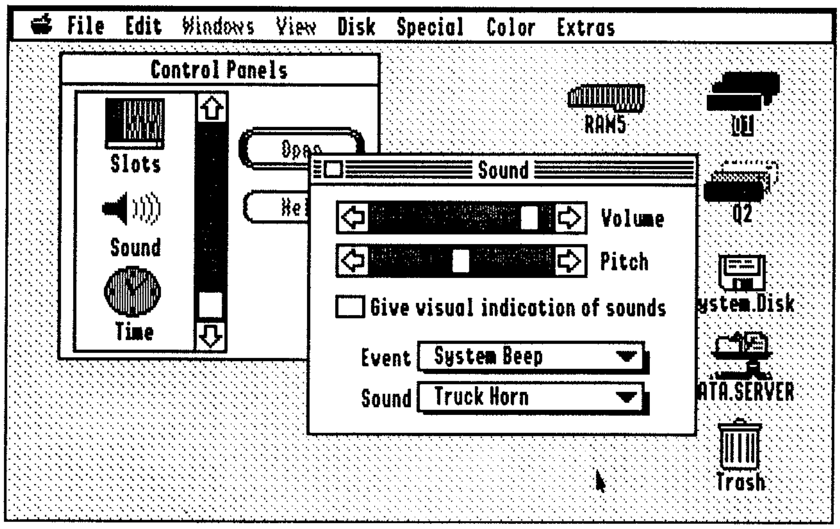


Figure 9L—Sound Control Panel

Most of the Events you can assign sounds to are self-explanatory. For example, if you assign a sound to the "Can't click there" event, you'll hear that sound whenever you click outside an alert or dialog window, because you're not supposed to click there. The Whoosh Open and Whoosh Closed sounds are played when a window "zooms" open or closed. The sound you assign to the "You Have Mail" event will probably never be heard—we don't know of any electronic mail packages for the Apple IIGS.

To assign a sound to an event, first choose the event you want to assign a sound to from the Event pop-up menu. Then choose the sound from the Sound pop-up menu. The Sound pop-up menu also allows you to assign silence to any event, or to assign the regular "bonk" beep to the event (the "Standard Beep" in the Sound menu). If an event is "Not assigned," the IIGS will either be silent or use whatever sound you've assigned to the System Beep, depending on the event. (For example, if "Can't click there" isn't assigned to any sound, then the System Beep sound will be used.)

Note!

The "Pitch" scroll bar only affects the Standard "bonk" Beep. It does not affect any of the other sounds in the Sound pop-up menu. Additionally, the pitch and volume are global—you can't assign different pitches and volumes to different events. (To do this, check out the Sonics module in Q Labs' Signature package.)

Time

The Time control panel (Figure 9M) sets the IIGS's clock and date/time formats, plus reveals the current time and date—even how many days have passed this year and how many are left.

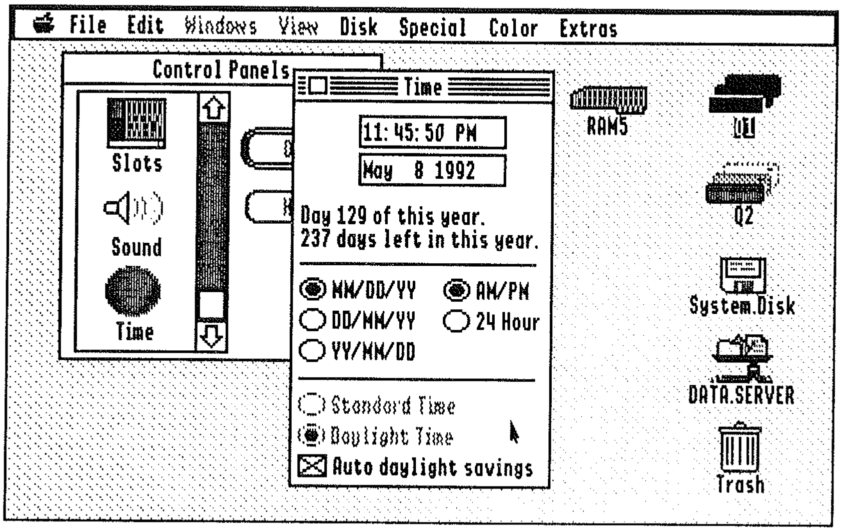


Figure 9M—Time Control Panel

To set the time or date, click the part you want to change (for example, the hour or the seconds) and use the tiny arrows that appear to adjust the number (or month, or AM/PM) up and down.

Shortcut

You can use the Tab key to move from field to field in the time and date. The up and down arrow keys scroll through the valid settings of the fields. You can type new values directly into the number fields (for example, to set the hour to 11 just type "11") and the AM/PM field (press "A" or "P").

The radio buttons below the clock settings are used to select the IIGS's default time and date display formats. Most Americans are used to seeing the month first, but in other countries the day comes first. You can also view the date with the year first, then the month and the day. The time can be displayed in AM/PM or 24-hour formats. These settings are used by most IIGS programs that display the date or time.

The Time control panel can also take care of Daylight Savings Time for you. If you live in an area that goes into Daylight Savings Time half the year, check the "Auto daylight savings" box and you'll never have to worry about springing your IIGS clock ahead or falling back—at least until the government changes when DST starts and ends again. If you live in an area which doesn't use Daylight Savings Time, or if you want to control your IIGS clock manually, uncheck the "Auto" box and click the appropriate radio button for standard or daylight time.

▲ By The Way...

If you believe in Santa Claus, click the text that tells you how many days are left in the year.

Chapter Ten

WORKING WITH OTHER PROGRAMS

10

About The Other Programs

In addition to the parts of System 6 we've talked about so far, the System 6 package also includes several applications that work with the System Software. Since these programs generally do something useful from a system standpoint, we call them utilities. These applications are:

- *Installer:* Installs and de-installs System 6 updates
- *Advanced Disk Utility (ADU):* Formats and partitions SCSI hard drives
- *Teach:* A simple word processor for reading documentation files and writing things
- *Archiver*: For making backup copies of disks and files
- *SynthLab:* A nifty music and synthesis program
- *Apple Bowl:* Okay, so it's not so useful. As games go, it's not even particularly state-of-the-art. For sheer nostalgia value, though, it's a winner.

The System 6 Bonus Pack adds:

- The Bonus Pack itself—a collection of DAs, Finder Extensions, clip arts, sounds, and other good stuff
- *HyperSound:* Converts System 6 "resource" sounds to HyperStudio-type sounds
- *FlashBoot:* A RAM Disk management utility
- *ShrinkIt:* A data compression/archival utility
- A couple other nifty things

The Installer

We already introduced you to the Installer way back in Chapter 2, but some further verbiage is in order, since we only touched the surface of the Installer's capabilities.

The Installer has two modes—"Easy Update" and "Customized" installation. You use the "Easy Update" most of the time; you use "Customized" installation when you want to install more than the "standard" configuration of system software (or when you want to remove part of the system software you're not using). We'll cover each mode separately—they're almost like two separate programs.

Easy Update

The Installer's Easy Update screen (Figure 10A) is a non-threatening dialog that allows you to select the disk you want to install the update on and begin installation, all by clicking buttons. The disk that will be updated is indicated by the message "Click 'Easy Update' to install Apple IIGS System Software on the disk 'Q1'"—or whatever the disk name might be. The buttons have the following functions:

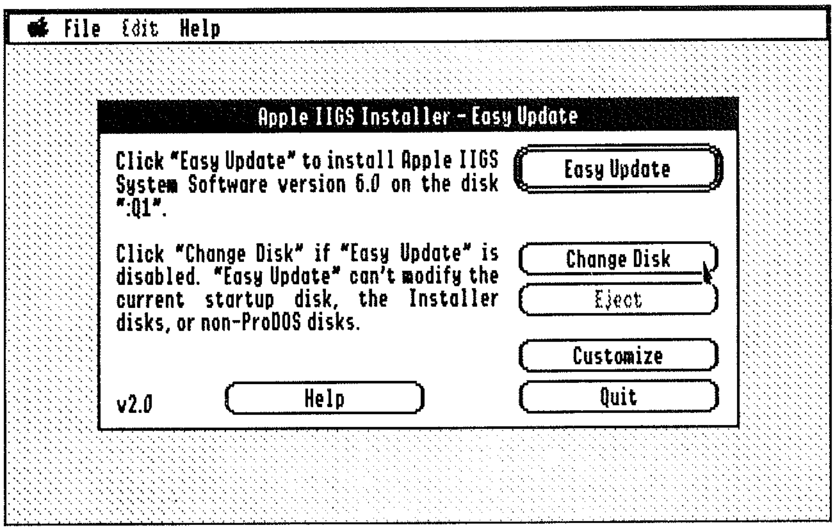


Figure 10A—The Installer's "Easy Update" Screen

- "Easy Update"—begins installation on the selected disk
- "Change Disk"—changes to the next disk connected to your IIGS
- "Eject"—ejects the chosen disk if the disk can be ejected
- "Customize"—switches to customized installation
- "Quit"—quits the Installer
- "Help"—displays a summary of the Installer's purpose

Additionally, the menu bar remains active—you can select the following menu items:

- "About" (menu)—displays credits for the Installer
- "Erase disk" (File menu)—erases the selected disk (you might occasionally want to erase an old disk before beginning installation)
- "Quit" (file menu)—quits the Installer
- Help menu—allows you to select from a variety of helpful hints

The "Easy Update" procedure generally installs the following system software on a drive that contains no existing system software (the software installed may vary based on your system configuration and the size of the disk you're installing to):

- System 6 and the Finder
- The Control Panels desk accessory
- The General, RAM, Printer, Slots, and SetStart control panels
- Drivers for Apple 5.25" and 3.5" drives—but *no* printer drivers
- Drivers for a SCSI Hard Drive connected to an Apple SCSI Card
- A File System Translator to access System 6 to access ProDOS disks
- Standard fonts (unless space is limited; then minimum fonts are installed)

If you perform an "Easy Update" on a disk that already contains system software (such as a hard drive that has System 5 installed on it), the Installer updates *all the currently installed system software* to new versions and deletes any files which are no longer needed.

▲ By The Way...

We recommend "Easy Update" as a good starting point for most System 6 installations. The Installer's help screens recommend a different installation procedure for hard drives (the "System 6: Hard Disk" script in custom installation, to be exact), but we don't recommend that method because it installs the Special Aids tools for disabled users, which most users won't need and which are incompatible with many popular programs (for example, AppleWorks GS).

Custom Installation

The Installer's "Custom" installation screen (Figure 10B) allows you to select various system updates from a scrolling list just by clicking the mouse. You can select multiple updates by holding down the ⌘ key while clicking the second and subsequent updates (the same maneuver unselects an update if you selected one accidentally). We've described each of these updates in some detail in Chapter 2; we won't repeat that information here.

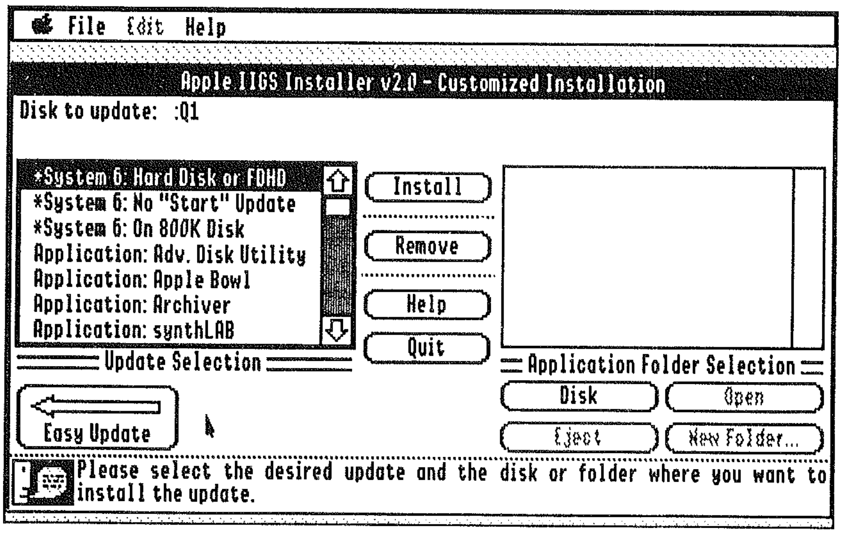


Figure 10B—The Installer's "Customized Installation" Screen

If you've selected an application (for example, Teach) for installation, the right side of the screen will become a Standard Save dialog, complete with "Open" and "New Folder" buttons and a pop-up folder menu, so you can tell the Installer where to put the application(s).

▲ **By The Way...**

The SynthLab application, since it consists of multiple files and folders, does not allow you to select a folder for installation. Instead, SynthLab is always installed in a folder called SynthLab in the main directory of the disk you've selected for installation.

As with the "Easy Install" screen, the "Customized Installation" window is controlled by clicking on buttons. The buttons are:

- "Install"—Installs the selected update(s)
- "Remove"—Removes the selected update(s)
- "Help"—Displays information about the selected update.
- "Quit"—Quits the Installer
- "Easy Update"—Returns to the "Easy Update" screen
- "Disk"—Chooses the next disk for installation
- "Eject"—Ejects a disk if it's removable
- "Open"—Opens a folder (accessible if you're installing one or more applications)
- "New Folder"—Creates and opens a new folder (accessible if you're installing one or more applications)

The menus remain active here, too—see the "Easy Install" section for details on each.

▲ **By The Way...**

You can write your own "scripts" to tell the Installer how to install software you develop. Explaining how to do that is, however, beyond the scope of this book. Consult IIGS Technical Note #64 (available from most user groups, online services, and via the Apple Programmers' and Developers' Association) for more details.

Advanced Disk Utility (ADU)

The Advanced Disk Utility (ADU) are useful mainly for preparing hard drives to store information; while it will work with other kinds of disks, most of the functions it performs can be performed just as well by the Finder if you're only dealing with floppies.

If you have only one hard drive and want to use ADU on it, you'll need to boot from a copy of the 3.5" System Disk.

The main ADU screen (Figure 10C) is the "launch pad" for the entire program. From here, you can perform the following functions by clicking on buttons:

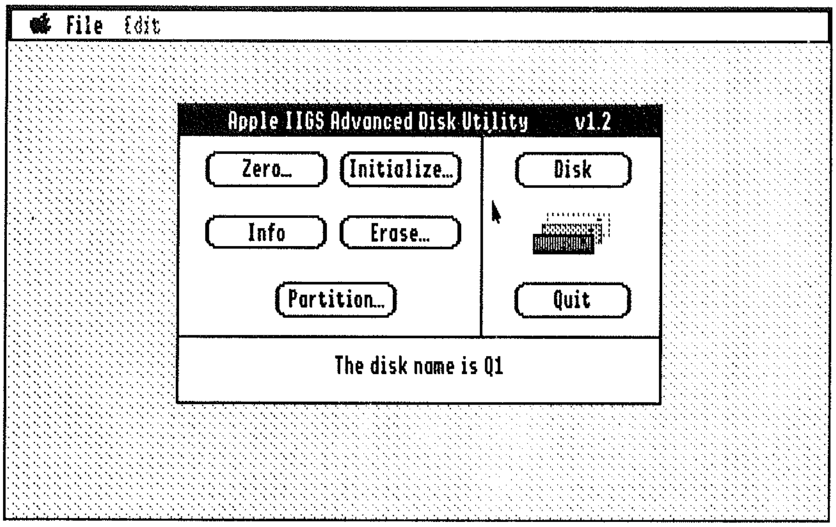


Figure 10C—The Advanced Disk Utility

- *Disk:* Chooses the disk on which to perform an operation.
- *Quit:* Quits the Advanced Disk Utility.
- *Initialize:*Prepares the disk to receive data. If your disk was a parking lot in which cars (files) can be parked, initializing it would be equivalent to painting lines and numbers on the pavement to divide the lot into parking spaces.
- *Zero:* Zeroing a disk erases all the data on it, including the operating system and the directory itself. After a disk has been zeroed, it must be initialized before any data can be stored on it. Zeroing a disk can take some time; it's equivalent to re-paving the parking lot.
- *Erase:* Erasing a disk removes all the cars from the parking lot, but leaves the lines and numbers in place by writing a new directory.
- *Partition:* Divides your parking lot into smaller parking lots. Each lot can hold different kinds of cars (files). Additionally, since ProDOS parking lots can have a maximum of thirty-two million spaces, you partition large hard drives into smaller ones so that you can use the whole drive.
- *Info:* Displays information about your parking lot, er, disk.

Warning!

The Initialize, Zero, and Partition operations erase all the data on the disk you use them on!

Partitioning A New Hard Drive

Usually you'll use ADU to prepare a new hard drive you've just received. (Most hard drive retailers do this for you, but you may come across a drive that isn't partitioned..) Here's how:

- Plan your partitions. We suggest making as many 32 megabyte partitions as you can, and assigning the remainder of the space to a spare partition. Another approach is to divide it equally—this works well on a 100 megabyte drive, which divides nicely into four 25 meg partitions (or three 32 meg partitions and one 4 meg partition). You could also make an HFS partition larger than 32 meg if you frequently use the hard drive on a Macintosh.
- Click the "Disk" button repeatedly until the drive you want to initialize is selected. (This will probably be the one without a name, as evidenced by the message "Uninitialized or no disk in drive" in place of the disk name.)
- Click the "Partition" button. The partition screen (Figure 10D) will appear. At first, the entire drive will be assigned to a single partition called "UNTITLED1."

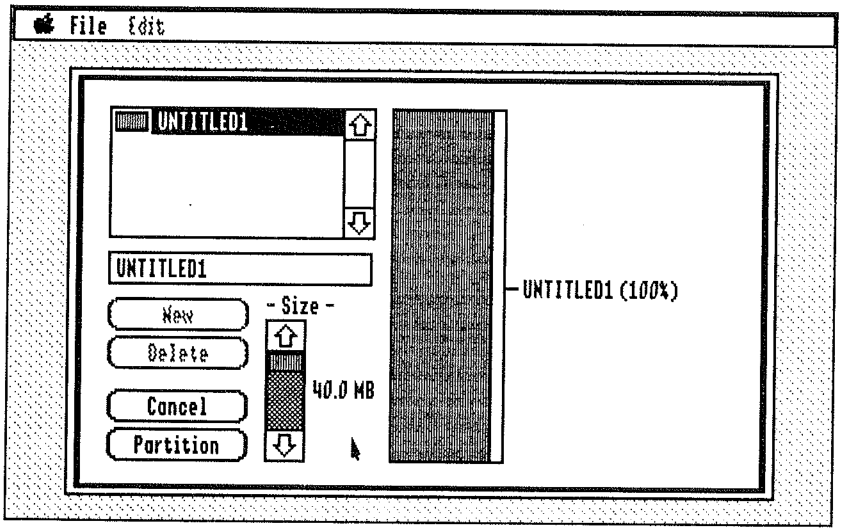


Figure 10D—Advanced Disk Utility: Partitioning A Hard Drive

- Type the name of the first partition. If you're not too creative, you can name your hard drive partitions Hard1, Hard2, Hard3, and so forth. Or you can name them after the Seven Dwarfs or the Seven Deadly Sins. Whatever floats your boat. Just make sure that the name conforms to regular ProDOS naming conventions.
- Set the partition's size using the Size scroll bar. The bar graph to the right of the screen displays the relative size of each partition.

- Press Return to confirm the name and size you specified.

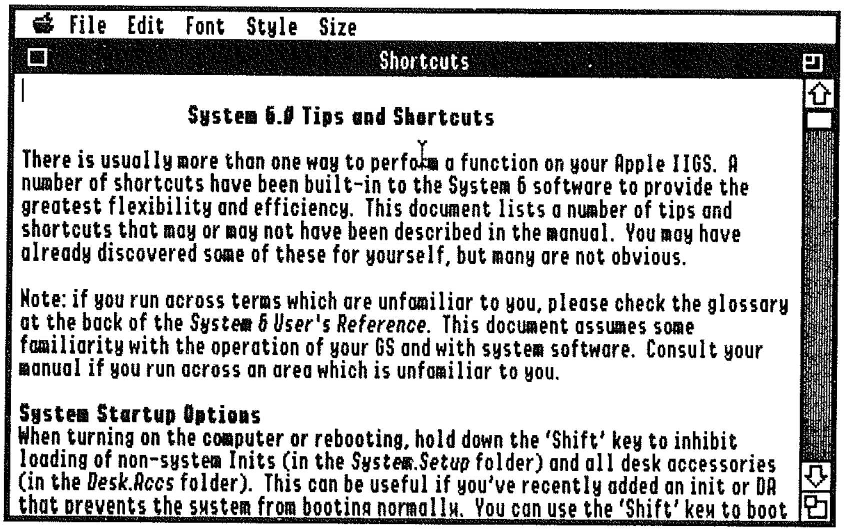
- Click New to create the next partition. Name it and set its size. If you try to assign more than the total amount of unassigned space, ADU will display a message to that effect and assign the rest of the space on the disk to the partition. Continue to create and assign new partitions until you've created all the partitions you want.

- Finally, make any last-minute changes to partition names and sizes. To change a partition's name or size, just click the partition's name in the partition list and make the changes to its name or size. If you have already allocated all the space on the drive and want to make a partition larger, you must first make one of the other partitions smaller.

▲ By The Way...

If you partition a drive that's already been partitioned, you'll come directly to the above screen to change the partition sizes.

- Now click the "Partition" button. Each partition will now be initialized; select its operating system (usually ProDOS is OK) from the standard Initialize dialog. Quit to the Finder and your newly partitioned hard drive should appear!



System 6.0 Tips and Shortcuts

There is usually more than one way to perform a function on your Apple IIGS. A number of shortcuts have been built-in to the System 6 software to provide the greatest flexibility and efficiency. This document lists a number of tips and shortcuts that may or may not have been described in the manual. You may have already discovered some of these for yourself, but many are not obvious.

Note: if you run across terms which are unfamiliar to you, please check the glossary at the back of the *System 6 User's Reference*. This document assumes some familiarity with the operation of your GS and with system software. Consult your manual if you run across an area which is unfamiliar to you.

System Startup Options
When turning on the computer or rebooting, hold down the 'Shift' key to inhibit loading of non-system Inits (in the *System.Setup* folder) and all desk accessories (in the *Desk.Accs* folder). This can be useful if you've recently added an init or DA that prevents the system from booting normally. You can use the 'Shift' key to boot

Figure 10E—Teach, System 6's Mini Word Processor

Teach

Teach (Figure 10E) is a simple word processor included with System 6. It includes the basic features you need to browse through files and to write and print short notes, although a full-featured word processor like Graphic Writer III or AppleWorks GS is better for larger projects. It supports as many simultaneously open files as you have memory for.

The windows are all standard IIGS windows that can be moved, resized, and scrolled. The regular text-edit actions are available for changing the text. The standard File options are all available, as well. To apply a font, first select the text, then select the desired font from the Fonts menu. Same with Style and Size. If you've been following along in our discussion of the Desktop, there's nothing here you haven't seen before.

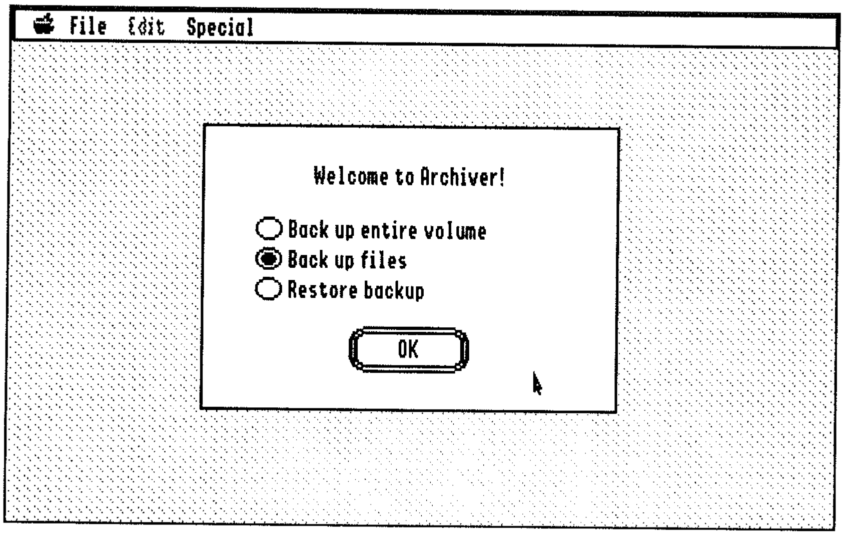


Figure 10F—The Archiver: Opening Screen

Archiver

The Archiver, too, is straightforward. It's a simple program for backing up the contents of your hard drive to floppies (or to a tape drive), or for backing up the contents of floppies into files on a hard drive. For this section, we're going to assume you're familiar with backing up your hard drive in some way and the basic concepts of backups.

The Archiver's opening screen (Figure 10F) allows you to select whether you want to back up an entire volume (as would be the case when making a backup of a 3.5" disk into a file on a hard drive) or the files on a volume (usually the case when making a backup of a hard drive—the Archiver has the capability to back up only the files that have changed after a certain date, making hard drive backups much more bearable). Or you can restore a backup you've already made (for example, when recreating your hard drive's contents after a crash).

Back Up Entire Volume

Backing up an entire volume is the simplest backup operation. The Volume Backup screen (Figure 10G) lets you select the volume to be backed up (via the "Volume" button), and whether you want to back up to a device or a file. If you want to back up an entire hard drive onto a set of floppy disks, select the "Device" radio button and click the "Device" button to select the drive that you'll put the disks into. If you'll be backing up a floppy disk into a file on a hard drive, click the "File" radio button and then click the "File..." button to create and name the file with a Standard Open dialog.

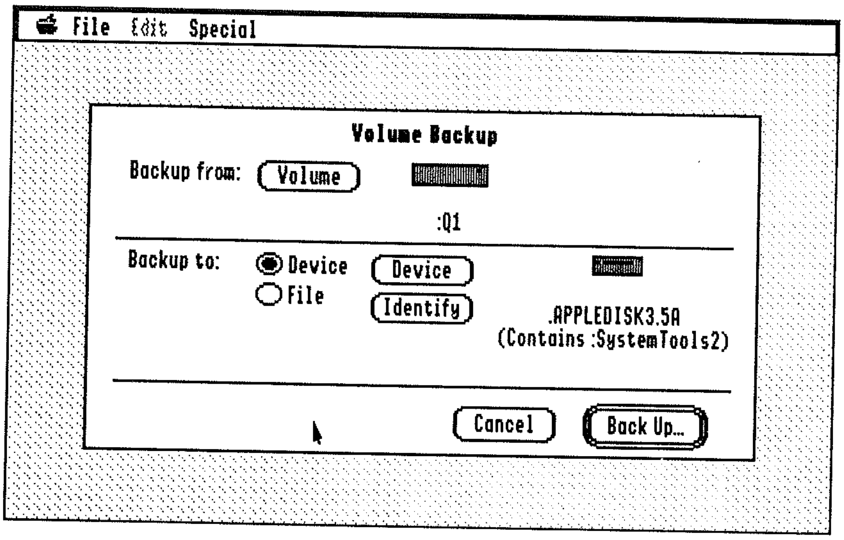


Figure 10G—Archiver Volume Backup Screen

When you've chosen those options, click "Back Up." On the next screen, enter a comment for this set of backup disks and click "Begin." And the operation will, in fact, begin. The Archiver will prompt you to switch disks when necessary, and will automatically initialize blank disks when they're inserted.

The "Back Up Entire Volume" operation backs up *everything* on the source disk on a block-by-block basis. It doesn't care about files, folders, or anything else. Thus, when you restore a volume backup, you must restore it to a volume of the same size, and the restoration operation will erase the existing contents of the volume.

Back Up Files

"Back Up Files" is more flexible. Selecting that brings you to the Selected Files Backup screen (Figure 10H). As before, you tell Archiver what volume you want to back up from, and the device (or file) you want to back up to. Usually, you'll select a 3.5" drive or SCSI tape drive, though you can also back up to a file on another hard drive.

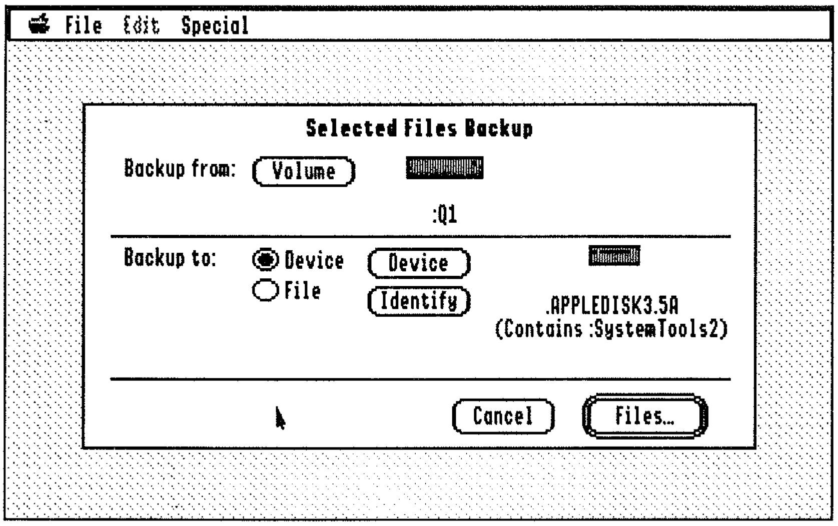


Figure 10H—Archiver File Backup Screen

Click "Files" and the Archiver will ask you if you want to use a saved file list. (The Archiver can remember a list of files you regularly back up, then back those up automatically.) For now, click "No." This brings you (in a few seconds) to the file selection screen (Figure 10J).

Here's where you tell the Archiver which files you want to back up. Initially, all the files on the volume you've chosen will be marked for backup. To mark or unmark a file or folder for backup, click the file's name and click "Mark" or "Unmark."

A gray folder indicates that some or all of the files in the folder are marked. You can double-click a folder to open it—the contents of the folder will be displayed, indented, below it. Each level of folder indents the listing further to the right. This way, you can mark and unmark individual files for backup, no matter how deep in the folder dungeon they are. Double-click the folder again to close it and hide its contents.

You can choose "Select All" from the Special menu, then click "Mark" or "Unmark" to mark or unmark all files for backup. (Marking a folder automatically marks all of its contents).

The "Auto" button is one of the Archiver's most powerful features. Click it and the Auto-Mark screen (Figure 10K) appears. Using a fill-in-the-blanks approach, you can tell the Archiver to mark (or unmark) certain files. To back up only the files that have changed since your last backup, you might first unmark all files, then use Auto-Mark to mark all the files whose modification ("mod") date is on or after your last backup date. You can also use this feature to mark all the files whose names meet certain criteria.

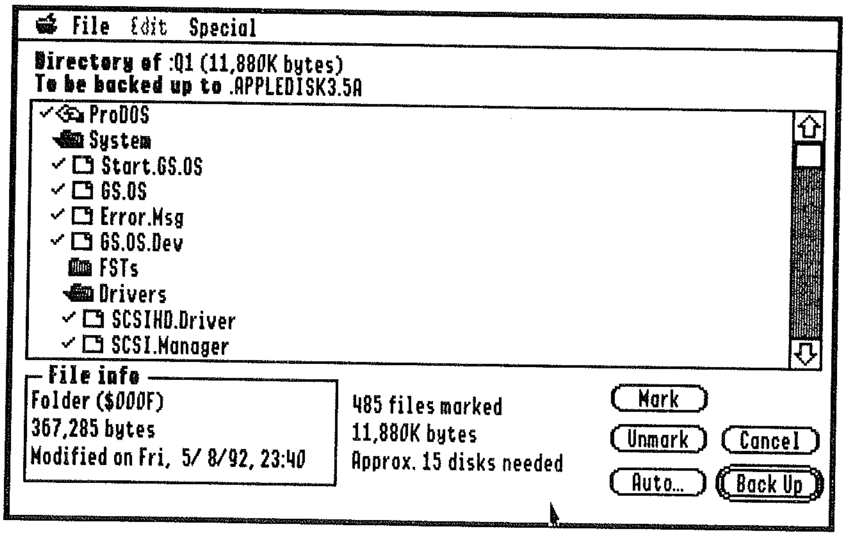


Figure 10J—Archiver File Selection Dialog

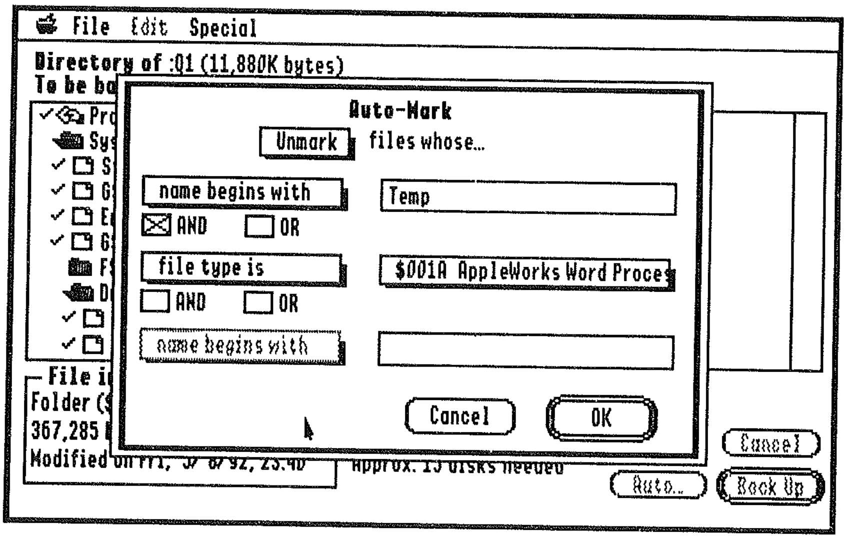


Figure 10K—Archiver Auto-Mark Dialog

You can use "and" and "or" logic to mark or unmark all files whose names start with the letter "A" *and also* were modified since the last backup date ("and" option), or to mark all the files containing "Data" and "AppleWorks" in one swoop ("or" option). Fill in this dialog then click "OK" to mark or unmark files according to the criteria you selected. The Archiver even goes into all your folders and applies your tests there.

Click "Back Up" and you'll have a chance to enter a name for your backup, just as before. Click "Begin" and you'll get started. As before, the Archiver will tell you when to switch disks.

Restore Backup

When you need to restore a backup (which we hope you never do—may your hard drive always run smoothly), use this option (Figure 10L). Tell the Archiver which kind of backup you want to restore from (File or Device) and identify the file or device. The Archiver will tell you the comment you entered when you made the backup.

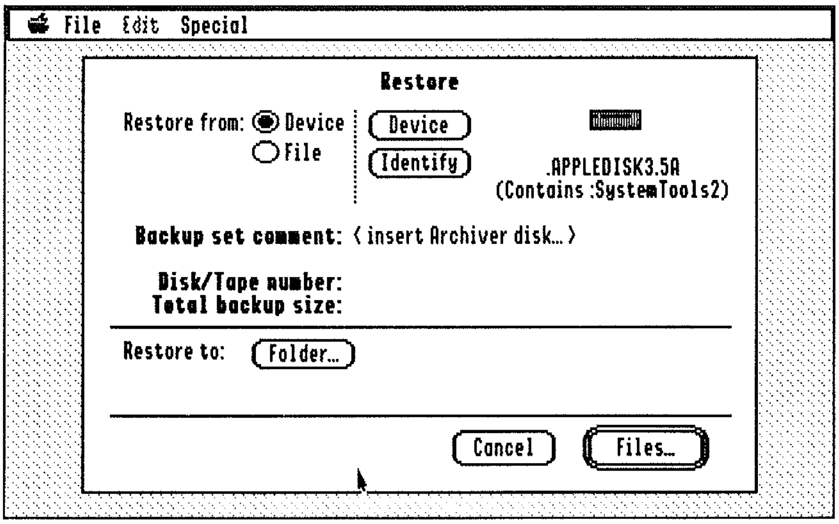


Figure 10L—Archiver Restore Screen

Click "Files" to select which files you'll be restoring (this only works when you're restoring a file-by-file backup). The file selection screen for this operation is much the same as when you backed up the files originally.

Click "Folder" to tell the Archiver where to put the restored files (again, not available with volume backups).

Click "Begin" and the restore will begin. Follow the prompts and insert the disks when requested.

Save File List

The "Save File List" menu option (on the File menu) allows you to save a list of the files you've marked with the Back Up Files option. If you often back up the same set of files, select this option from the pull-down menu and give the set a name. Then you can use that set when you back up by file the next time.

Preferences

The Special menu's "Preference" option displays the Archiver Preferences dialog (Figure 10M).

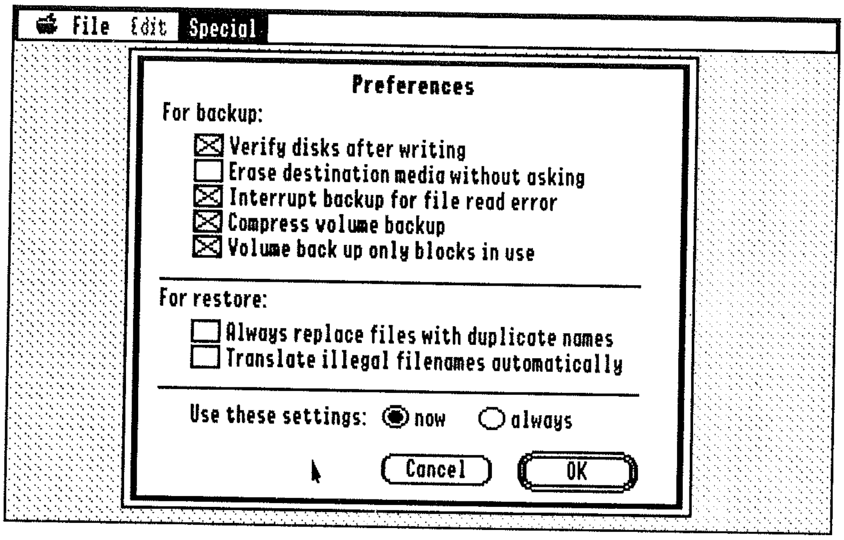


Figure 10M—Archiver Preferences Dialog

- *Verify disks after writing*—makes sure that the archived data was written correctly after filling a backup disk. We recommend leaving this option checked.
- *Erase destination media without asking*—allows the Archiver to erase the disks you're using for your backup without asking first, even if there's already something on them.
- *Interrupt backup for file read error*—If you have errors on your hard drive but want to continue backing up your hard drive in the event of an error (to back up all the good files), uncheck this option.
- *Compress volume backup*—Uses data compression to make backing up a volume smaller.
- *Volume back up only blocks in use*—For most disks, you can leave this option checked. If the volume's bitmap is damaged or invalid, you can uncheck this option to back up all the blocks on the disk, even if they're not marked as being used by files.

- *Always replace files with duplicate names*—Check this option if you always want to replace files already on the disk with older versions from the backup.
- *Translate illegal filenames automatically*—Forces the names of the restored files to fit the format of the disk you're restoring to without asking.

- *Use these settings now*—Uses the settings for the current Archiver session but doesn't save them for the next session.
- *Use these settings always*—Saves the settings for future Archiver sessions and uses them for the current session as well.

SynthLab

SynthLab (Figure 10N) is a nifty program for recording and playing music on your IIGS. However, its sound design and sequencing facilities require some knowledge of synthesis and MIDI technology, and it's not our goal to teach you these things. Most users will use it only for playing back pre-recorded songs (several are included), and that's all we're going to explain. If you're adventurous, you may be able to figure some of the rest of it out by yourself. (If you're a programmer or musician, contact the Apple Programmers' and Developers' association for more involved documentation on SynthLab and the MidiSynth toolset.)

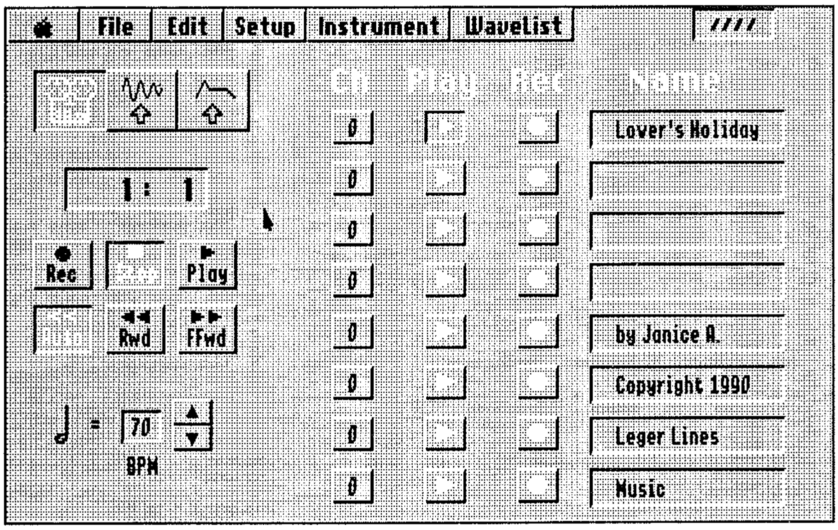


Figure 10N—SynthLab Music & Synthesis Utility

SynthLab is a psuedo-desktop program. It looks like a desktop program and it's controlled by the mouse, but it behaves somewhat differently from most desktop programs. For example, the menu bar doesn't pull down the menus in sequence if you drag across the menu bar; you have to release one menu before choosing another. The menu, too, doesn't list all your NDAs. Why Apple itself would release a program that doesn't follow the most basic of Desktop user interface guidelines is beyond us, but there it is.

The "Open Sequence" option on the File menu is where most users will start. The Standard Open dialog opens up to the Seq.And.Instr folder inside the SynthLab folder, but you can also maneuver it to other folders. To play a sequence you've loaded, click the "Play" button (the one with the rightward-pointing arrow). Simple enough, right?

To quit the program, choose "Quit" from the File menu.

Apple Bowl

As we've mentioned before, this program isn't really good for much but nostalgia value. Well, it is a pretty good little bowling game—if you don't mind the fact it was written about ten years ago for the Apple II (not the II+, not the IIe—the Apple II). The program, in fact, appeared on an early Apple II system disk. (Or maybe it was a cassette tape—we can't remember that far back.)

The program is self-explanatory, or at least self-documenting, and it's also a good demonstration of how you can be running a IIGS program and still not be using the Desktop. Have fun.

▲ **By The Way...**

Apple Bowl was originally written in Integer BASIC (remember that?). Apple engineers used The Byte Works' ORCA/Integer Basic compiler to convert it to a standalone IIGS program.

The Bonus Pack

We've already explained how to use some of the Bonus Pack's components—the desk accessories and finder extensions, in the chapter on Desk Accessories. The Fonts, Sounds, and Icons, when installed in the proper places in your System folder, are automatically available to you—the fonts appear in any application that allows you to choose fonts for your text, and the sounds appear automatically in the Sounds control panel. The icons show up in the Finder—we've included icons for many popular applications and documents.

▲ **By The Way...**

We've included TrueType versions of five of the Bonus Fonts. If you have WestCode's Pointless software, unpack the file Truetypes.SHK from the Bonus Fonts 2 disk to your TrueType font directory and configure Pointless to use them. If you don't have Pointless, don't worry; the same fonts are also included as standard IIGS fonts.

The clip art files are standard Super High Resolution (SHR) graphics screens and can be loaded by programs that use graphics such as Platinum Paint, HyperStudio, HyperCard IIGS, and AppleWorks GS. Some are 640-mode graphics, which have fine detail but are limited in color, and others are 320-mode graphics, which have plenty of color but not as much fine detail. Check your graphics programs to see which files they'll deal with.

HyperSound

The sounds we've included are in System 6 format, known among the technically-minded as "rSound" (or resource sound) format. System 6 and HyperCard IIGS can use these sounds fine. But HyperStudio and many other popular programs can't use these sounds. So we've included an exclusive program called HyperSound, which converts "rSound" sounds to "HyperStudio" sounds, which HyperStudio (naturally) and other programs can use. The program is on the Bonus Stuff disk and can be launched from the Finder.

The program converts the sounds "in place"—meaning that when the program has finished running, you won't have rSounds anymore. Therefore, you shouldn't use HyperSound to convert the sounds in your startup disk's Sound folder. Make another copy of the sounds you want to use with HyperStudio (or other programs) first so you don't lose the original rSounds.

HyperSound is quite simple to operate. There aren't any menus because the program only does one thing. Just use the multi-file Open dialog to choose the file(s) you want to select (hold down the ⌘ key to select the second and subsequent files, or select all the files in a directory by clicking the first one, then shift-clicking the last one). Then click the "Accept" button and HyperSound will convert the sounds.

FlashBoot

FlashBoot allows you to use the IIGS's RAM Disk to store your most frequently used programs and switch between them quickly, and to also streamline loading and backing up your RAM Disk each time you use it. It's ideal for users without hard drives.

☞ **Note!**

Instead of including a separate (but nearly empty) FlashBoot disk, we included the FlashBoot program on the Bonus Stuff disk. Start up via the Bonus Stuff disk to use FlashBoot.

We've included a separate manual for this program. The manual was written before System 6 was released, but hardly anything has changed. Use the Installer to perform an Easy Install on the RAM Disk after preparing it with FlashBoot, then install your favorite programs as described in the FlashBoot manual.

☞ **Note!**

Pardon the blatant sales ploy here, but did you know you can now buy a whopping four megabytes of RAM for your computer—making it into a perfect machine for FlashBoot—for less than $200 from Quality Computers?

ShrinkIt

You've already used ShrinkIt (on the Bonus Stuff disk) to install the Bonus Pack. It's a good general-purpose data compression utility and is frequently used to archive files for transfer via modem. By compressing your least frequently used files with ShrinkIt, you can keep them handy on the hard drive but free a good deal of the space they usually take up. You'll have to unpack them again before you can use them, of course, but ShrinkIt is good for storing old files you won't need to reference very option.

A documentation file for ShrinkIt is included on the Bonus Stuff disk. Use Teach to read it. We highly recommend this utility for all Apple IIGS users!

☞ **Note!**

ShrinkIt author Andy Nicholas now works for Apple. In fact, he wrote the System 6 Finder (along with Dave Lyons, whose Memory Bar NDA is included in the Bonus Pack).

The Other Nifty Things

Gee, as if the Bonus Pack isn't enough—we've also included ZZCopy (which you used to back up your original System 6 disks). (You did back up your original System 6 disks, right?) Documentation for the program can be printed or viewed from within the program—click the blank panel in the middle of the bottom of the screen, then click the printer or monitor icon (next to "Documentation") to read or view the instructions.

The ZZCopy disk had some extra space, so we included a few other programs—an icon editor, a font editor, and a few other things. Insert the ZZCopy disk at the Finder to access these files. Some of these programs are shareware, which means that you should send the author of the program a fee if you find the program useful. These extra programs *are not* strictly part of the Bonus Pack, and buying the Bonus Pack does *not* entitle you to use the shareware programs without paying the required fee.

All these programs have included documentation files, which can be opened and printed with Teach. The ZZCopy disk *only* can be distributed to your friends; please don't make illegitimate copies of the other disks in the Bonus Pack.

☞ **Note!**

We are acting only as a distributor for the programs on the ZZCopy disk (including ZZCopy itself). We do not guarantee that these programs work for any particular purpose, nor do we provide technical support for the programs. And we'd like to remind you once again—pay for the shareware programs if you use them. You have a legal and moral obligation to do so, plus you'll be supporting the authors so they can keep producing shareware programs and thus keep the Apple II market interesting.